Neuroengineering

Course Notes

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Part I
Prof. Cincotti

1 EEG

1.1 Introduction to EEG

The electroencephalogram, or EEG, is a non-invasive technique used to record electrical potential differences from the scalp. These potentials reflect, in an indirect and spatially filtered way, the activity of large populations of neurons, especially cortical neurons.

Historically, non-invasive human EEG became important in the 1920s and 1930s thanks to Hans Berger, who observed a large and persistent oscillation around 10 Hz, later called the *alpha rhythm*. Later, Adrian and Matthews proposed the idea that some brain rhythms may appear when a cortical area is relatively idle, that is, when it is not strongly engaged in a specific task.

Author note.

The idea of an "idling rhythm" should not be interpreted as the brain being switched off. Rather, it means that an area may show a more synchronized oscillatory pattern when it is not actively processing a specific stimulus or task. When that area becomes functionally engaged, the rhythm often decreases in power.

1.2 Spontaneous Brain Rhythms

EEG contains spontaneous oscillatory activity. The main frequency bands are usually classified as follows:

Band	**Frequency range**	**General meaning**
Delta	< 3.5 Hz	Slow activity
Theta	4–7.5 Hz	Slow/intermediate activity
Alpha	8–13 Hz	Around 10 Hz rhythm
Beta	14–30 Hz	Faster activity
Gamma	> 30 Hz	High-frequency activity

The alpha rhythm is typically recorded over occipital regions and is strongly related to the visual system. It is often prominent when the eyes are closed and decreases when visual processing increases.

The mu rhythm lies in the same frequency band as alpha, approximately 8–13 Hz, but it has a different functional and spatial meaning. It is recorded mainly over central sensorimotor regions and is related to motor activity, motor preparation, and motor imagery. Its waveform is often more arc-shaped than sinusoidal.

Author note.

A very important point is that frequency alone is not enough to identify a rhythm. Alpha and mu can both be around 10 Hz. To distinguish them, one must consider location and functional modulation. A 10 Hz rhythm over occipital electrodes that changes with eyes open or closed is likely alpha. A 10 Hz rhythm over central electrodes that changes with movement or motor imagery is likely mu.

1.3 Evoked and Event-Related Potentials

Besides spontaneous activity, EEG can show responses that are time-locked to external or internal events. A response directly driven by a sensory stimulus is often called an *evoked potential* (EP). The more general term *event-related potential* (ERP) refers to EEG changes related to external stimuli, motor actions, cognitive events, or task-related processes.

ERP analysis became possible with the use of computers, because EEG responses to single events are usually very small and hidden inside the spontaneous EEG. By repeating the same event many times and averaging the EEG epochs aligned to that event, the signal-to-noise ratio improves.

If the EEG response is similar across trials and the background EEG behaves like random noise, averaging N trials improves the signal-to-noise ratio approximately by:

$$\sqrt{N}.$$

Author note.

The key idea is simple: the event-related response is consistent across trials, while the unrelated EEG activity varies more randomly. When we average many trials, the consistent part remains and the random part is reduced.

1.4 EEG Electrodes and Transducers

EEG electrodes are transducers that interface the biological tissue with the electronic recording system. In the body, electrical currents are mainly carried by ions, whereas in wires and amplifiers they are carried by electrons. The electrode is therefore the interface between ionic conduction and electronic conduction.

Common EEG electrodes are made of silver/silver-chloride, Ag/AgCl. These electrodes are useful because they are relatively non-polarizable and allow stable recordings, including slowly changing potentials.

In EEG terminology, electrode impedance usually refers to the quality of contact between the electrode and the scalp, through conductive gel or another conductive medium.

Author note.

A useful way to remember this is: the electrode does not "create" the EEG signal. It allows the recording system to access potential differences generated by biological activity.

1.5 EEG Amplifiers

EEG signals are very small, usually in the range of microvolts. Therefore, EEG amplifiers must satisfy strict requirements:

- low noise;
- high gain;
- high input impedance;
- high common-mode rejection ratio (CMRR).

EEG amplifiers are typically differential amplifiers. Ideally, they amplify the difference between two input potentials:

$$V_{\text{out}} \approx A(V_+ - V_-),$$

where A is the gain of the amplifier.

A ground electrode is also attached to the subject. It provides a common electrical reference for the amplifier system. Its position can be chosen rather freely, provided that it does not introduce a large common-mode signal.

1.6 Common-Mode Rejection Ratio

The common-mode rejection ratio, CMRR, measures how well an amplifier rejects signals that are common to both inputs. A good EEG amplifier should amplify the differential signal of interest while suppressing common-mode disturbances such as power-line noise.

The CMRR is defined as the ratio between differential gain and common-mode gain:

$$\mathrm{CMRR} = 20\log_{10}\left(\frac{\text{differential gain}}{\text{common-mode gain}}\right).$$

High CMRR values, often around 100 dB, indicate better rejection of common-mode noise.

Author note.

The amplifier does not want to amplify everything that reaches the electrodes. It should amplify what is different between the two inputs and suppress what is common to both. This works well only if the recording system is balanced.

1.7 Input Impedance and Electrode-Scalp Impedance

A crucial distinction must be made between:

- the input impedance of the amplifier;
- the contact impedance between electrode and scalp.

The input impedance of the amplifier should be very high, typically in the order of hundreds of megaohms. This prevents the amplifier from loading or attenuating the biological signal.

The electrode-scalp impedance should instead be low and similar across electrodes. Traditional EEG guidelines suggest keeping electrode impedance below approximately $5\,\mathrm{k}\Omega$, although modern high-input-impedance amplifiers allow this requirement to be somewhat relaxed.

If electrode impedances are too high or strongly unbalanced, two problems may occur:

- the biological potential may be attenuated;
- the common-mode rejection of the whole recording system may worsen.

No exam / technical intuition.

The green-background slides explain this using the voltage-divider model. The intuition is that if the contact impedance becomes comparable to the amplifier input impedance, part of the biological signal is lost across the contact impedance instead of being measured by the amplifier.

Moreover, if the two electrodes have very different impedances, common-mode noise no longer enters the two amplifier inputs in the same way. This asymmetry reduces the effectiveness of CMRR.

1.8 Standard Electrode Positions: The 10–20 System

The international 10–20 system defines standard EEG electrode positions using anatomical landmarks:

- nasion;
- inion;
- left and right preauricular points;
- vertex.

The name 10–20 comes from the fact that electrode distances are defined as 10% or 20% of the distances between reference points on the skull.

Electrode labels indicate scalp regions:

Label	**Meaning**
F	Frontal
C	Central
P	Parietal
O	Occipital
T	Temporal
z	Midline

Odd numbers indicate the left hemisphere, while even numbers indicate the right hemisphere. For example, C3 is a central left electrode, C4 is a central right electrode, and Cz is located on the midline near the vertex.

Author note.

For motor imagery and mu rhythm studies, C3, Cz, and C4 are especially important because they lie over sensorimotor areas.

1.9 Reference Electrode and Re-Referencing

Each EEG channel is a potential difference. Therefore, EEG always depends on the chosen reference. Common reference positions include earlobes, mastoids, and the nose. However, these sites are not truly inactive.

The position of the reference electrode can strongly influence the shape and amplitude of EEG waveforms. For this reason, reference choice is not a minor technical detail.

One common approach is the common average reference (CAR), where the instantaneous average of all channels is subtracted from each channel. In ideal theoretical conditions, this would approximate a reference at infinity. In practice, these ideal conditions are never perfectly satisfied.

Author note.

A useful sentence to remember is: the EEG reference changes what we see. The same neural activity may appear with different amplitude or waveform depending on the reference used.

1.10 Monopolar and Bipolar Montages

In a monopolar montage, all channels are measured with respect to a common reference. This is useful for studying spatial potential distributions over the scalp.

In a bipolar montage, each channel is the difference between two nearby electrodes. Bipolar montages are widely used in clinical EEG because they help highlight local changes and phase reversals.

> **Author note.**
> The monopolar montage answers the question: "What is the potential at each electrode relative to a reference?" The bipolar montage answers: "How does the potential change between nearby electrodes?"

1.11 Good Practices in EEG Data Collection

Good EEG data require careful experimental preparation. Recordings should be performed with alert and cooperative subjects in conditions as free as possible from artifacts and noise.

The recording session should be divided into smaller runs or blocks, separated by breaks. Subjects should be reminded to:

- relax their muscles;
- minimize blinking;
- avoid head and body movements;
- avoid jaw clenching, swallowing, and facial tension.

The experimenter should continuously monitor the EEG online to detect artifacts or technical problems. Accurate logging is also essential: each run should be documented, and relevant metadata or time markers should be saved with the data.

> **Author note.**
> EEG cannot be fully fixed after acquisition. Signal processing can reduce artifacts, but bad acquisition, missing markers, or unstable electrodes can seriously compromise the analysis.

1.12 EEG Artifacts

An EEG artifact is any potential difference due to an extracerebral source. Artifacts may be biological or technical.

Biological artifacts include:

- eye movements and blinks;
- muscle activity;
- cardiac activity;
- sweating;
- body and head movements.

Technical artifacts include:

- power-line noise;
- bad electrode contact;
- electrode detachment;
- electrical interference from external devices;
- ADC saturation.

Artifact prevention is preferable to post-hoc artifact removal.

Author note.
Many artifacts have amplitudes much larger than the actual EEG signal. Once they contaminate the recording, they can sometimes be attenuated, but this may also remove useful neural information.

1.12.1 Eye-Related Artifacts

The eye behaves like an electrical dipole: the cornea is more positive than the retina. When the eye moves, this dipole moves inside the volume conductor and produces large potentials detectable by EEG electrodes.

Eye movement artifacts can be around 0.5 mV, much larger than ordinary EEG rhythms. Eye blinks produce large, slow, often monophasic deflections lasting around 200–400 ms. These artifacts are especially visible over frontal electrodes.

Author note.
If very large slow waves appear mainly over frontal channels, the first suspect should be blinks or eye movements, not necessarily brain activity.

1.12.2 Muscle Artifacts

Muscle activity, EMG, can contaminate EEG recordings. Muscle artifacts may range from hundreds of microvolts to millivolts and have a broad frequency spectrum, from tens of Hz to several kHz.

This is especially problematic because EMG overlaps with beta and gamma EEG bands. Common sources include jaw clenching, swallowing, tongue tension, forehead contraction, and neck muscle activity.

Author note.
If the EEG trace looks very fast, irregular, and high-frequency, especially over temporal or frontal areas, EMG contamination should be suspected.

1.12.3 Cardiac Artifacts

The heart generates electrical potentials recorded as ECG or EKG. These can contaminate EEG, especially when the reference electrode is not placed on the head.

Another cardiac-related artifact is the ballistocardiogram, caused by small electrode movements due to nearby blood vessels pulsating. This artifact can sometimes be reduced by slightly moving the electrode.

1.12.4 Sweating and Movement Artifacts

Sweating produces electrodermal responses, which are slow and high-amplitude potentials, typically below 0.5 Hz.

Head and body movements may also produce artifacts. These may come from muscle activity or from mechanical displacement of the charged double layer at the electrode-skin interface.

Author note.
Slow artifacts are not harmless. They can strongly affect baseline correction and ERP analysis, especially when studying slow components.

1.13 Power-Line Noise

Power-line noise arises from capacitive coupling with alternating current from electrical power supplies. In Europe, it is typically centered around 50 Hz; in other countries, it may be 60 Hz. Harmonics may also appear.

A notch filter can selectively attenuate 50 Hz noise. However, it should be used carefully if the frequency band of interest is close to the rejected frequency.

Author note.
For mu and beta analysis, typically 8–30 Hz, a 50 Hz notch filter is usually not too problematic. For gamma-band analysis, however, it may remove or distort useful information.

1.14 Offline EEG Inspection

Before any advanced processing, raw EEG data should be visually inspected. Bad channels or bad epochs must be identified and removed when necessary.

The principle is:

garbage in, garbage out.

However, one should avoid rejecting too much data. The decision depends on the specific analysis and on which artifacts are relevant for the research question.

Author note.
There is no universal cleaning pipeline. If the goal is to study frontal ERPs, blinks are a major problem. If the goal is to study occipital alpha, isolated frontal blinks may be less damaging, but baseline shifts and movement artifacts may still be critical.

1.15 Segmentation and Averaging

For ERP or EP analysis, the continuous EEG recording is segmented into epochs. Each epoch usually includes:

- a pre-stimulus or pre-event baseline period;
- a post-stimulus or post-event period.

The epochs are aligned to the event of interest and averaged sample by sample. This synchronized averaging preserves the event-related response and reduces unrelated spontaneous EEG activity.

ERP peaks are usually characterized by:

- amplitude, measured relative to the pre-stimulus baseline;
- latency, measured relative to the stimulus or event onset.

ERP components are often named using polarity and nominal latency. For example:

- P100: positive peak around 100 ms;
- N170: negative peak around 170 ms;
- P300: positive peak around 300 ms.

Author note.

The number in the component name is nominal, not exact. A negative peak at 108 ms may still be called N100 if it corresponds to the physiological N100 component.

1.16 Stimulus Timing: SOA, ISI, and ITI

In EEG experiments, timing is crucial. The main timing terms are:

Term	Meaning
SOA	Time between the onset of one stimulus and the onset of the next
ISI	Time between the end of one stimulus and the onset of the next
ITI	Time between the beginning of one trial and the beginning of the next

If the stimulus duration is long, SOA and ISI can be very different. ERP amplitude and shape may depend strongly on stimulus timing. Long-latency components are especially sensitive to fast stimulus repetition.

Author note.

There is a trade-off: more trials improve the signal-to-noise ratio, but if stimuli are too close in time, responses may overlap or decrease in amplitude.

1.17 Evoked and Induced Activity

Evoked activity is both time-locked and phase-locked to the stimulus. Therefore, it can be revealed by conventional time-locked averaging.

Induced activity is related to the event but not phase-locked. It may occur at a similar frequency after each stimulus, but with variable phase or latency. Because of this jitter, it may disappear in the average waveform.

To study induced activity, one can analyze amplitude or power in a specific frequency band, for example by filtering the signal and then rectifying or squaring it before averaging.

Author note.

ERP analysis is good for activity that is aligned in time and phase. If the brain responds with a change in oscillatory power but with variable phase, time-frequency analysis or ERD/ERS analysis is more appropriate.

1.18 Event-Related Desynchronization and Synchronization

Event-related desynchronization and synchronization, ERD/ERS, quantify changes in EEG power relative to a baseline period.

The typical procedure is:

1. band-pass filter the EEG in the frequency band of interest;
2. square the signal to estimate power;
3. average across trials;
4. smooth the result using short time windows;
5. compute the percentage change relative to baseline.

The general formula is:

$$\text{ERD/ERS}(\%) = \frac{P_{\text{task}} - P_{\text{baseline}}}{P_{\text{baseline}}} \cdot 100.$$

If the value is negative, there is ERD. If the value is positive, there is ERS.

- **ERD**: decrease in power relative to baseline.
- **ERS**: increase in power relative to baseline.

In motor imagery, ERD in the mu band over C3 during right-hand imagery suggests involvement of the left sensorimotor cortex.

Author note.
ERD does not simply mean "less brain activity". In sensorimotor EEG, ERD often means that the synchronized rhythm is disrupted because the cortical area is functionally engaged.

1.19 Time-Frequency Analysis

Time-frequency analysis is used to observe how the spectral content of EEG changes over time. It is especially useful for studying induced activity and ERD/ERS.

There is always a trade-off between time resolution and frequency resolution. Short time windows provide good temporal resolution but poor frequency resolution. Long time windows provide good frequency resolution but poor temporal resolution.

Author note.
This trade-off is one of the main practical limitations of time-frequency EEG analysis. You cannot arbitrarily obtain perfect timing and perfect frequency precision at the same time.

1.20 Live EEG Demonstration

The live EEG demonstration used a wireless 32-channel amplifier and active Ag/AgCl electrodes. These electrodes include on-board impedance converters, producing a low-impedance signal through the wire and therefore reducing susceptibility to interference.

The setup requires symmetric cap placement based on anatomical landmarks such as nasion, inion, preauricular points, and vertex. Impedances must be monitored. A scientific target is usually below $5\,\text{k}\Omega$, while a more relaxed demonstration target may be below $25\,\text{k}\Omega$.

The demonstration also illustrates the practical effect of filters:

- high-pass filters reduce slow drifts;
- low-pass filters reduce high-frequency components;
- notch filters attenuate power-line noise at 50 Hz.

Author note.
The demonstration connects the theoretical concepts to practical observation. Removing the notch filter may reveal 50 Hz noise; asking the subject to clench the jaw may reveal EMG; changing the high-pass filter may strongly affect slow baseline fluctuations.

1.21 Core Exam Takeaways

The most important points to remember are:

1. EEG measures potential differences, not absolute brain activity.
2. Alpha and mu rhythms may have similar frequencies but different locations and meanings.
3. ERP analysis relies on time-locked averaging to improve signal-to-noise ratio.
4. Evoked activity is phase-locked; induced activity is not necessarily phase-locked.
5. Amplifiers need high input impedance, while electrode-scalp impedance should be low and balanced.
6. High CMRR is essential to reject common-mode disturbances such as power-line noise.
7. The reference electrode is not neutral and can influence EEG waveform shape and amplitude.
8. Artifacts can be biological or technical and are often larger than the brain signal.
9. ERD/ERS measure percentage changes in band power relative to baseline.
10. Good data acquisition is more important than aggressive post-processing.

1.22 True Statements

This section collects the true statements explicitly provided in Prof. Cincotti's EEG lectures. They are useful as compact exam-oriented facts.

1.22.1 Spontaneous EEG Rhythms and Event-Related Responses

1. The alpha rhythm is an oscillatory component of the spontaneous EEG.
2. The frequency of oscillation of the alpha rhythm is around 10 Hz.
3. The amplitude of the EEG in the alpha band decreases when the generating region of the cerebral cortex becomes engaged in a functional task, such as a visual or motor task.
4. The proper visual alpha rhythm is generated in the occipital lobe of the cerebral cortex.
5. The mu rhythm is an oscillatory component of the spontaneous EEG whose frequency lies in the alpha band.
6. The mu rhythm is generated in the central regions of the cerebral cortex.
7. The oscillations of the mu rhythm are more arc-shaped rather than resembling a regular sine wave.
8. The amplitude of the alpha rhythm is not constant, but rather waxes and wanes with irregular periods, with changes often occurring after intervals in the order of one second.
9. The alpha band of the EEG is conventionally limited between 8 and 13 Hz.
10. The delta and theta frequency bands identify frequencies lower than those in the alpha band.
11. The beta and gamma frequency bands identify frequencies higher than those in the alpha band.

12. The alpha rhythm can be observed by filtering the spontaneous EEG signal using a narrow-band filter, with cutoff frequencies approximately at 8 and 13 Hz.

13. Evoked potentials are deflections of the EEG signal following the presentation of a sensory input.

14. Event-related potentials include evoked potentials, as well as EEG responses to motor or cognitive events.

1.22.2 Electrodes, Impedance, Amplifiers, and Montages

1. EEG electrodes made of silver with a layer of silver chloride, Ag/AgCl, are non-polarizable and therefore allow the recording of extremely slow-changing potentials.

2. In EEG terminology, impedance is a measure of the quality of the contact between electrode and scalp, through the conductive gel.

3. Contact impedance of the electrodes is measured in kilo-ohms, kΩ, and must be measured using an alternating current.

4. Gold electrodes are polarizable; therefore, they should not be used to measure slowly changing potentials.

5. The Common-Mode Rejection Ratio, CMRR, of a differential amplifier characterizes the ratio between the gain of the difference of potential between input electrodes and the gain of the common potential with respect to ground.

6. The CMRR is usually expressed in decibels, dB, and high values characterize better amplifiers.

7. The advantage of a high-CMRR amplifier is that it suppresses common-mode disturbances, such as power-line noise at 50 Hz.

8. The measurement of a single EEG signal requires three electrodes: two electrodes as inputs to the differential amplifier and one electrode to provide the ground potential.

9. The input impedance of a biosignal amplifier must be many orders of magnitude higher than the contact impedance of the electrodes. It is usual to have input impedances in the order of $10^8\,\Omega$.

10. EEG guidelines suggest keeping the electrodes' contact impedance below $5\,\mathrm{k}\Omega$. The use of modern amplifiers with high input impedance allows this requirement to be relaxed.

11. One reason for keeping the electrodes' contact impedance much lower than the amplifier's input impedance is that the resulting voltage divider would otherwise reduce the measured potential.

12. The difference between contact impedances of electrodes should be small compared to the input difference of the differential amplifier; otherwise, the resulting unbalance compromises its common-mode rejection capability.

13. The electrode labels of the International 10–20 System use the first letter of the four lobes of the cerebral cortex, Frontal, Parietal, Occipital, Temporal, plus "C" for central electrodes over the central sulcus.

14. In the electrode labels of the International 10–20 System, odd numbers designate electrodes on the left side and even numbers designate electrodes on the right side.

15. The International 10–20 System for EEG electrode placement takes its name from the fact that the distance between adjacent electrodes is 10% or 20% of the distance between pairs of reference points on the skull, such as nasion and inion or the preauricular points.

16. In monopolar EEG recordings, a single reference potential is used for all recorded channels. The reference electrode is placed on scalp positions assumed to be far from the electrical sources of interest, such as the earlobes.

17. The position of the reference electrode can strongly influence the shape and amplitude of EEG potentials. The profile of scalp topographies, disregarding the actual potential value, is not influenced.

18. In bipolar EEG recordings, each channel is the difference of potential between two adjacent electrodes. This montage is mostly used in clinical settings to highlight features of interest during visual inspection of the waveforms.

19. EEG signals recorded in monopolar configuration can be re-referenced to the Common Average Reference, CAR, by subtracting from each channel the instantaneous average of all channels. In ideal conditions, this would approximate taking the reference potential at infinity.

1.22.3 EEG Artifacts

1. An EEG artifact is a potential difference due to sources outside the brain.

2. Artifacts can have biological origin, such as eyes, muscles, and heart; they can be due to external electromagnetic generators, such as power supply or engines; or they can be due to events affecting the recording setup, such as electrode movements, loss of contact, or saturation of the ADC.

3. Artifacts can be partially attenuated or removed through signal processing during data analysis, but it is always preferable to make all efforts to prevent them during signal acquisition.

4. The eye is more positive in its frontal part, the cornea, than in its posterior part, the retina. Therefore, eye movements can generate large EOG artifacts on the EEG, of the order of $500\,\mu$V.

5. During an eyeblink, the eyelid shortcuts the positive potential of the external surface of the eye, causing positive EEG deflections with amplitudes of several hundreds of microvolts and durations of a few hundreds of milliseconds.

6. A sudden upward or downward movement of the eyes generates a positive or negative deflection of EEG potentials, respectively, on frontal EEG channels.

7. A sudden movement of the eyes to the right or left generates a positive or negative deflection of EEG potentials, respectively, on the EEG channel F8.

8. The electrical activity of muscles, EMG, has spectral content starting at frequencies of about 20 Hz and above, thus affecting the beta and gamma bands of the EEG signal.

9. The amplitude of the EMG generated by muscles in the head can be ten times higher than the spontaneous EEG signal, thus in the order of 1 mV.

10. EMG artifacts can easily appear in EEG recordings unless subjects are specifically instructed by the experimenter on how to relax their face, tongue, and throat muscles.

11. Heart activity can contaminate EEG recordings because an electrocardiographic artifact, ECG or EKG, can directly affect the potentials, especially if the reference electrode is not placed on the head.

12. Heart activity can also contaminate EEG recordings because a ballistocardiographic artifact is indirectly generated by the pulse of a blood vessel causing movements of a nearby electrode.

13. Sweating can affect EEG, causing slow-changing and high-amplitude artifacts below 0.5 Hz, up to a few millivolts.

14. Power-line noise is an artifact caused by capacitive coupling between conductors carrying alternating current, typically at 50 Hz, and the recording setup, including the subject.

15. Power-line noise affects a very narrow frequency band of the recorded signal around 50 Hz or 60 Hz, depending on the power-line frequency. Other frequency bands can be affected at multiples, typically odd multiples, of 50 or 60 Hz.

16. Power-line noise is accentuated by asymmetries in the recording electrode pairs, such as impedance differences and cable path differences, because asymmetries prevent the noise from being rejected by the amplifier's common-mode rejection capabilities.

17. Notch filters effectively remove power-line noise because they selectively reject the narrow band affected by the artifact, preserving almost entirely the useful signal.

18. Movement of the subject's head may produce slow artifacts on the EEG recording, whose waveform is closely related to the time course of the movement.

19. Since movement-related potentials can originate from the mechanical displacement of the charged double layer at the electrode interface, these artifacts are less pronounced when non-polarizable electrodes are used.

1.22.4 ERP, EP, ERD/ERS, and Time-Frequency Analysis

1. As a preliminary step to EEG data analysis, the following data can be discarded: one or more channels, if they are extensively contaminated by artifacts; and all time intervals, or epochs, in which artifacts appear. Both strategies can be applied to the same dataset.

2. The estimation of event-related or evoked potentials, ERPs and EPs, requires the acquisition of numerous repetitions, typically tens or hundreds, of the stimulus or event that evoked or induced the potential.

3. When recording ERPs or EPs, whose peak amplitudes range from a few microvolts down to a fraction of a microvolt, the spontaneous EEG, whose amplitude is of tens of microvolts, is to be considered noise that completely masks the EPs or ERPs on the recorded waveform.

4. In ERP analysis, the averaging procedure consists of segmenting epochs, or trials, with fixed duration from the raw recording, each aligned to a repetition of the event, and then performing a synchronized average, that is, averaging all corresponding samples sharing the same latency across the set of trials.

5. Synchronized averaging of N trials preserves the amplitude of the ERP and reduces the amplitude of the background spontaneous EEG activity by a factor $\sqrt{N}$, under commonly verified hypotheses.

6. The amplitude of ERPs is measured with respect to a baseline epoch, usually preceding the stimulus, in which the amplitude is assumed to be zero.

7. The latency of an ERP peak is measured with respect to the relevant event, usually the presentation of a stimulus, rather than with respect to the beginning of the waveform.
8. In an ERP, peaks are named with a leading P if the peak has positive polarity and with a leading N if the peak has negative polarity.
9. In an ERP, peaks are often named with a trailing number indicating the nominal latency in milliseconds. In older conventions, the trailing number represents the order of the peak within the ERP.
10. A negative peak in an ERP recorded on a specific subject with a latency of 108 ms may still be named N100 if it matches the physiological phenomenon of the nominal N100 component.
11. The Stimulus Onset Asynchrony, SOA, measures the time interval between the onsets of two successive stimuli in a train. If each trial contains only one stimulus, it is equivalent to the Inter-Trial Interval, ITI.
12. The Inter-Stimulus Interval, ISI, measures the time interval between the end of one stimulus and the beginning of the following one. It equals the SOA minus the stimulus duration.
13. In an ERP, the response to a stimulus has reduced amplitude when the SOA is too short. Long-latency components of the ERP are especially sensitive to this decrease.
14. Brain activity in response to a stimulus can be phase-locked to the event, meaning that the whole time course, including positive and negative peaks, has the same latency in every repetition. This activity is called evoked.
15. Brain activity in response to a stimulus can be non-phase-locked, meaning that it shows variable latency, or jitter, at each repetition. This activity is called induced.
16. The averaging procedure can reliably uncover components of an ERP corresponding to evoked brain activity.
17. Induced activity is often examined by analyzing the envelope of the EEG in a relevant frequency band, that is, by rectifying or squaring the pass-band filtered trials before averaging them.
18. In EEG terminology, synchronization and desynchronization are synonymous with an increase and decrease, respectively, of waveform amplitude and thus power.
19. Event-Related Desynchronization/Synchronization, ERD/S, quantifies changes of EEG power relative to a baseline period, expressed as percentage change. ERD/S is usually evaluated over a whole range of relevant latencies with respect to the event.
20. Computation of ERD/S from a set of N EEG trials requires the following steps: band-pass filtering in the relevant frequency band; squaring each sample; synchronized averaging across trials; smoothing by averaging samples within short adjacent time windows covering the whole time course; and computing the relative percentage change with respect to the average value in a baseline period.
21. An alternative algorithm to evaluate ERD/S uses rectification instead of squaring, thus estimating the amplitude of the signal rather than its power.
22. Time-frequency analysis allows the analysis and visualization of changes in the spectral components of a signal over time.
23. Time resolution and frequency resolution cannot be freely chosen: the higher one is, the lower the other becomes.

2 Signal Processing

2.1 Role of Signal Processing in Biomedical Measurements

A typical biomedical measurement system starts from physiological or external energy and converts it into a signal that can be processed by a computer. The general pipeline is:

$$\text{physiological phenomenon} \to \text{transducer} \to \text{amplifier} \to \text{analog processing} \to \text{ADC} \to \text{digital signal processing.}$$

In neuroengineering, signal processing has two main roles:

- preprocessing or conditioning, that is, improving the quality of the signal before further analysis;
- feature extraction, that is, transforming raw signals into meaningful numerical descriptors.

Examples of preprocessing include filtering, artifact attenuation, and signal segmentation. Examples of feature extraction include estimating signal power in a frequency band, computing the PSD, extracting ERP amplitudes, or measuring EMG amplitude features.

Author note.
The key idea is that biomedical signals are rarely used directly in their raw form. We usually process them to remove irrelevant components and extract features that are more directly related to the physiological question.

2.2 Analog-to-Digital Conversion

Analog-to-Digital Conversion, ADC, transforms a continuous analog signal into a digital sequence of numbers. It consists of two main operations:

$$\text{ADC} = \text{sampling} + \text{quantization.}$$

Sampling discretizes time. Quantization discretizes amplitude.

- **Sampling**: the signal is measured at discrete time instants.
- **Quantization**: each measured amplitude is approximated to the nearest allowed digital level.

Author note.
A useful way to remember the difference is: sampling decides *when* we measure; quantization decides *how precisely* we represent the amplitude.

2.3 Quantization and Quantization Noise

Quantization approximates each analog value to one of a finite number of digital levels. If the ADC uses N_{bits} bits, the number of available quantization levels is approximately:

$$2^{N_{\text{bits}}}.$$

The width of a quantization interval is called LSB, Least Significant Bit. If the input range of the ADC is R, then approximately:

$$LSB = \frac{R}{2^{N_{\text{bits}}}}.$$

Quantization introduces an error, called quantization noise. Under standard assumptions, this noise can be modeled as additive, uniformly distributed, zero-mean noise. Its standard deviation is:

$$\sigma_q = \frac{LSB}{\sqrt{12}}.$$

Increasing the number of bits reduces the quantization interval and therefore reduces the quantization noise.

Author note.
Quantization noise is not the same as physiological noise. It is an artifact introduced by the digital representation itself. A good ADC should make quantization noise smaller than the intrinsic noise of the input signal.

2.4 Input Range, Precision, and Clipping

The ADC input range must be chosen carefully.

If the range is too large, the quantization interval becomes larger, increasing quantization noise. If the range is too small, large-amplitude components may exceed the ADC range and produce clipping or saturation.

$$\text{large range} \rightarrow \text{larger quantization error},$$

$$\text{small range} \rightarrow \text{higher risk of clipping}.$$

Author note.
This is a practical trade-off. You want the range to be large enough to avoid saturation, but not unnecessarily large, otherwise you waste amplitude resolution.

2.5 Sampling and the Nyquist-Shannon Theorem

The Nyquist-Shannon sampling theorem states that a continuous signal can be correctly sampled only if it does not contain frequency components above one half of the sampling frequency.

If the sampling frequency is f_s, the Nyquist frequency is:

$$f_N = \frac{f_s}{2}.$$

For proper sampling, the maximum frequency component of the signal must satisfy:

$$f_{\max} < f_N.$$

Equivalently:

$$f_s > 2f_{\max}.$$

Author note.
The theorem does not say that every signal sampled above twice the frequency of interest is

automatically usable. It assumes that frequencies above Nyquist have been removed before sampling.

2.6 Aliasing

Aliasing occurs when a signal contains frequency components above the Nyquist frequency before sampling. These high-frequency components are incorrectly represented as lower-frequency components in the sampled signal.

For a sinusoidal component with frequency:

$$f_0 \in (f_N, f_s),$$

the aliased frequency is:

$$f_{\text{alias}} = f_s - f_0.$$

Aliasing is dangerous because, after sampling, the false low-frequency component cannot be distinguished from a real low-frequency component.

Author note.
Aliasing is not just a loss of information. It is worse: high-frequency content is actively misrepresented as low-frequency content.

2.7 Anti-Aliasing Filtering

Aliasing can be prevented by applying an analog low-pass filter before the ADC. This filter is called an anti-aliasing filter.

Its role is to remove frequency components above the Nyquist frequency before sampling:

$$f_{\text{cutoff}} < f_N.$$

This must be done in the analog domain, before digitization.

Author note.
A digital filter applied after sampling cannot undo aliasing, because once the high-frequency component has been folded into the low-frequency range, it is already indistinguishable from genuine low-frequency content.

2.8 Analog Signal Conditioning

Real ADCs are limited by input range and maximum sampling frequency. Therefore, the analog signal should be conditioned before conversion so that:

- the signal range does not exceed the ADC input range;
- the signal bandwidth does not exceed what is needed for faithful representation;
- high-amplitude artifacts are attenuated when possible;
- frequencies above Nyquist are removed.

Analog filtering is therefore a key step before ADC.

2.9 Statistics, Probability, and Noise

Signals can be deterministic or stochastic.

A deterministic signal is fully described by an explicit rule or waveform. Examples include sine waves, square waves, and triangular waves.

A stochastic signal is described statistically. Its exact sample values cannot be predicted deterministically, but its statistical properties can be described.

Probability theory models random variables and random processes. Statistics describes empirical observations and estimates the parameters of probabilistic models.

Author note.

In signal processing, the word "sample" can be ambiguous. In statistics, a sample is a set of observations. In signal processing, a sample is one measurement of a signal at a specific time instant.

2.10 Basic Signal Measures

Some common measures for signal characterization are:

- mean;
- variance;
- standard deviation;
- root mean square, RMS;
- average rectified value, ARV.

The mean of a signal with samples x_i is:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i.$$

The variance is estimated as:

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2.$$

The standard deviation is:

$$s = \sqrt{s^2}.$$

The RMS is:

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}.$$

The ARV is:

$$ARV = \frac{1}{N}\sum_{i=1}^{N} |x_i|.$$

For zero-mean signals, RMS is closely related to standard deviation.

Author note.

RMS and ARV are often used when we care about signal amplitude but the signal oscillates around zero. This is particularly important for EMG, where raw positive and negative phases would cancel out if we used the mean.

2.11 White Noise and Gaussian Noise

White noise is a stochastic signal whose samples are uncorrelated. Knowing one sample does not help predict another sample.

In the frequency domain, ideal white noise has a flat spectrum, meaning it has the same power at all frequencies.

Gaussian noise is noise whose sample amplitude follows a normal distribution, often with zero mean.

$$x \sim \mathcal{N}(0, \sigma^2).$$

Author note.

Whiteness describes correlation across time. Gaussianity describes the distribution of amplitudes. A signal can be white without being Gaussian, and Gaussian without being perfectly white.

2.12 Central Limit Theorem and Averaging

The Central Limit Theorem states that the average of many independent and identically distributed random variables tends to a Gaussian distribution as the number of variables increases.

If N independent random variables have standard deviation σ, the standard deviation of their average is:

$$\sigma_{\text{avg}} = \frac{\sigma}{\sqrt{N}}.$$

This is why averaging reduces random noise.

For example, if N EEG trials contain only spontaneous EEG noise with RMS equal to σ, their synchronized average has RMS approximately:

$$RMS_{\text{avg}} = \frac{\sigma}{\sqrt{N}}.$$

Author note.

This is the mathematical reason why ERP averaging works: the event-related component is time-locked and remains, while unrelated random activity is reduced by averaging.

2.13 Fourier Analysis

Fourier analysis decomposes a signal into a sum of sine and cosine components. Each sinusoidal component carries power at a specific frequency.

This allows us to move from the time domain:

$$x(t)$$

to the frequency domain:

$$X(f).$$

The frequency domain representation tells us which frequencies are present in the signal and how much amplitude or power each frequency carries.

2.14 Discrete Fourier Transform

The Discrete Fourier Transform, DFT, transforms a time-limited and sampled signal into a discrete frequency-domain representation.

If a signal has N samples, its DFT contains N complex values. For a real-valued time-domain signal, the frequency spectrum is symmetric:

$$X(-f) = X^*(f).$$

For this reason, only the first half of the spectrum, up to the Nyquist frequency, is needed to characterize the signal.

The DFT represents each frequency component through:

- magnitude, or absolute value;
- phase, or angle.

> **Author note.**
> Magnitude tells us how much of a frequency is present. Phase tells us where that sinusoidal component is shifted in time.

2.15 Fast Fourier Transform

The Fast Fourier Transform, FFT, is an efficient implementation of the DFT. It is especially efficient when the number of samples is a power of two.

$$FFT = \text{efficient algorithm for computing the DFT.}$$

2.16 Frequency Resolution

If a signal has total duration $T_{\max}$, the distance between adjacent DFT frequency bins is:

$$\Delta f = \frac{1}{T_{\max}}.$$

Thus, longer signals provide better frequency resolution.

$$T_{\max} \uparrow \quad \Rightarrow \quad \Delta f \downarrow.$$

> **Author note.**
> Frequency resolution is determined by observation time, not directly by sampling frequency. Sampling frequency determines the maximum representable frequency; signal duration determines the spacing between frequency bins.

2.17 Zero-Padding

Zero-padding means adding samples with zero amplitude at the end of a signal before computing the DFT.

Zero-padding reduces the distance between adjacent displayed frequency samples, making the spectrum look smoother.

However, it does not add new information and does not truly improve the physical frequency resolution determined by the original signal duration.

> **Author note.**
> Zero-padding is interpolation in the frequency domain. It helps visualization and peak localization, but it does not create new data.

2.18 Windowing and Spectral Leakage

When we analyze a finite section of a longer signal, we are effectively multiplying the signal by a windowing function.

Using a rectangular window may introduce discontinuities at the edges of the selected segment. These discontinuities cause spectral leakage: energy from one frequency spreads into nearby frequencies.

Spectral leakage is characterized by:

- a main lobe;
- side lobes.

Tapered windows, such as Hamming or Blackman-Harris windows, reduce side lobes but widen the main lobe.

> **Author note.**
> There is a trade-off. A rectangular window is better when we need to separate close frequency components of similar power. A tapered window is better when leakage from a strong component could hide a weaker component at another frequency.

2.19 Periodogram and Power Spectral Density

The periodogram is the squared magnitude of the DFT of a time-limited signal. It estimates how signal power is distributed across frequency.

For stochastic signals, a single periodogram can have high variability. Therefore, one often estimates the Power Spectral Density, PSD, using averaging methods.

The PSD is a statistical description of signal power as a function of frequency.

2.20 Averaged Periodogram and Welch Method

The averaged periodogram method divides a long signal into shorter epochs. The spectrum of each epoch is computed and then spectra are averaged bin by bin.

Welch's method improves this approach by:

- splitting the signal into possibly overlapping epochs;
- applying a windowing function to each epoch;
- computing the spectrum of each windowed epoch;
- averaging spectra bin by bin.

This reduces the variance of the PSD estimate, but at the cost of lower frequency resolution.

Author note.

Welch's method is useful because biological signals are noisy. Averaging spectra makes the estimate more stable, but shorter windows mean worse frequency resolution.

2.21 Short-Time Fourier Transform and Spectrogram

The Short-Time Fourier Transform, STFT, estimates the spectrum of a signal in short moving time windows.

It is used to analyze non-stationary signals whose frequency content changes over time.

The spectrogram displays:

- time on the horizontal axis;
- frequency on the vertical axis;
- magnitude or power using color.

Author note.

STFT introduces the classical time-frequency trade-off. Short windows give good time resolution but poor frequency resolution. Long windows give good frequency resolution but poor time resolution.

2.22 Digital Filters

A digital filter allows some spectral components of a signal to pass almost unaltered while attenuating other components.

The four basic types are:

- low-pass filter;
- high-pass filter;
- band-pass filter;
- band-stop or band-reject filter.

A notch filter is a narrow band-stop filter, commonly used to attenuate power-line noise at 50 Hz, or 60 Hz in some countries.

2.23 Passband, Stopband, Transition Band, and Cutoff

The passband is the frequency interval in which the filter gain is close to 1.

The stopband is the frequency interval in which the filter gain is close to 0.

The transition band is the region between passband and stopband.

The cutoff frequency, or corner frequency, marks the limit of the passband. In many filter designs, the gain at cutoff is approximately:

$$\frac{1}{\sqrt{2}} \approx 0.71,$$

corresponding to:

$$-3\,\mathrm{dB}.$$

The roll-off is the slope of the frequency response in the transition band. A fast roll-off means a narrow transition band.

2.24 Good Filter Properties

Good filter properties include:

- fast roll-off;
- flat passband, with no or little ripple;
- strong stopband attenuation;
- minimal distortion of relevant signal components.

Stopband attenuation is better evaluated in decibels rather than on a linear gain scale, because small gains such as 0.001 and 0.0001 are difficult to distinguish visually in linear scale.

2.25 FIR and IIR Filters

Digital filters are classified into:

- FIR filters, Finite Impulse Response;
- IIR filters, Infinite Impulse Response.

The output of an FIR filter is a linear combination of input samples.

The output of an IIR filter depends on both input samples and previous output samples.

IIR filters are usually more efficient than FIR filters, meaning that a lower order is often sufficient to reach similar frequency-domain specifications.

However, FIR filters can be designed to have linear phase, while IIR filters generally cannot.

Author note.
Linear phase means that all frequency components are delayed by the same amount. This is important when preserving waveform shape matters, for example in ERP analysis.

No exam / technical intuition.
The additional filter draft material provides deeper examples of convolution, nonlinear phase effects, and extra demonstrations. These are useful for intuition, but they should be treated as supplementary unless explicitly included in the exam program.

2.26 Core Exam Takeaways

1. ADC consists of sampling and quantization.
2. Sampling discretizes time; quantization discretizes amplitude.
3. The Nyquist frequency is half of the sampling frequency.
4. Aliasing occurs when the signal contains components above Nyquist before sampling.
5. Aliasing must be prevented using analog anti-aliasing filtering before ADC.
6. Quantization noise depends on the quantization interval, LSB.
7. A large ADC range increases quantization error; a small range increases clipping risk.
8. Averaging N independent noise realizations reduces standard deviation by $\sqrt{N}$.
9. The DFT represents a sampled signal in the frequency domain.

10. For real signals, the spectrum is symmetric and the second half is redundant.

11. Frequency resolution is determined by the total observation time.

12. Zero-padding improves visual interpolation but does not add information.

13. Windowing controls spectral leakage.

14. Welch's method reduces PSD variance at the cost of spectral resolution.

15. Digital filters are low-pass, high-pass, band-pass, or band-stop.

16. FIR filters can have linear phase; IIR filters are usually more efficient.

2.27 True Statements

2.27.1 ADC, Sampling, Aliasing, and Quantization

1. Shannon's theorem, or sampling theorem, states that a continuous signal can be properly sampled only if it does not contain frequency components above one half of the sampling rate.

2. In Analog-to-Digital Conversion, ADC, the Nyquist frequency equals half of the sampling frequency.

3. The reconstruction of an analog signal from its sampled version is equivalent to the sum of a set of sinc() functions, one for each sample, each centered on the time of the respective sample, whose amplitude equals the sample value.

4. Aliasing occurs when an analog signal is sampled outside the conditions set by Shannon's theorem.

5. When a sinusoidal component with frequency $f_0 \in (f_{\text{Nyquist}}, f_{\text{sampling}})$ is sampled at frequency f_{sampling}, aliasing occurs and the reconstructed signal is a sinusoidal component at:
$$f_{\text{aliasing}} = f_{\text{sampling}} - f_0.$$

6. Aliasing can be prevented by applying an analog low-pass filter with cutoff frequency lower than f_{Nyquist} to the analog signal before it is converted.

7. Quantization is the approximation of the analog value of a sample to the nearest among the allowed quantization levels.

8. Quantization introduces noise whose amplitude is proportional to the width of the quantization interval:
$$\sigma_{\text{noise}} = \frac{1}{\sqrt{12}} LSB.$$

9. Quantization divides the input range of the ADC into approximately $2^{N_{\text{bits}}}$ intervals, where N_{bits} is the number of bits of the ADC.

10. Given a fixed number of bits of the ADC, choosing a large input range increases the quantization error, while choosing a small input range increases the chance that the signal is clipped.

11. Appropriate application of an analog filter before conversion may prevent saturation by removing high-amplitude artifacts in specific frequency bands.

2.27.2 Statistics, Probability, and Noise

1. A signal can be entirely deterministic, or it can be known only by means of its statistical properties.

2. Probability theory deals with the mathematical modeling of random variables and stochastic processes.

3. Statistics deals with the description of empirical observations and with the estimation of usually unknown parameters of mathematical models.

4. In statistics, a sample is a collection of observations drawn from a population.

5. In signal processing, a sample is a single measurement of a signal at a specific time point.

6. The Average Rectified Value, ARV, is obtained by summing the absolute values of all samples and dividing by the number of samples.

7. The Root Mean Square, RMS, is the square root of the average of the squared values of the samples of a signal.

8. The variance of a signal is estimated by summing the squared deviations of the sample values from the sample mean and dividing by $N - 1$.

9. The standard deviation of a signal is the square root of its variance.

10. In white noise, all samples are uncorrelated.

11. The frequency spectrum of white noise is flat.

12. In Gaussian noise, the probability density that a sample has a given amplitude follows the normal distribution, usually with zero mean.

13. The Central Limit Theorem states that the probability distribution of the average of N independent and identically distributed random variables tends to a normal distribution for N approaching infinity.

14. The probability distribution of the average of N independent and identically distributed Gaussian random variables is normal independently of the value of N.

15. Given N independent and identically distributed random variables with variance σ^2, the variance of their average is σ^2/N.

16. For independent and identically distributed random variables with standard deviation σ, the standard deviation of their average is $\sigma/\sqrt{N}$.

17. The synchronized average of N trials containing only spontaneous EEG with RMS equal to σ has RMS equal to $\sigma/\sqrt{N}$.

2.27.3 Fourier Transform, DFT, and Spectral Analysis

1. Signals can be decomposed into sums of sine waves, each carrying power at a single frequency.

2. The Discrete Fourier Transform associates the time-limited and sampled time-domain representation of a signal with its bandwidth-limited and sampled frequency-domain representation.

3. Given a sampled signal with N samples, its DFT contains N complex values.

4. For a real time-domain signal, the first half of the DFT is sufficient to completely define the signal.
5. The DFT represents the amplitude and initial phase of sine wave components of the signal.
6. Only components up to the Nyquist frequency are needed to characterize the signal, since the remaining part of the spectrum is a mirrored copy of the first half.
7. Fast Fourier Transform, FFT, is an efficient implementation of the DFT, especially for signals whose number of samples is a power of two.
8. The spectrum of a signal is often represented as magnitude and phase plots.
9. Spectral analysis is well suited to identifying narrowband useful signals in approximately white noise.
10. For a signal sampled with sampling interval $\Delta T_s = 1/f_s$, the spectrum has a frequency range of width f_s.
11. For a signal of total duration $T_{\max}$, adjacent DFT frequency bins are $\Delta f = 1/T_{\max}$ apart.
12. Zero-padding a signal reduces the distance between adjacent displayed samples of the spectrum, visually improving frequency resolution.
13. Spectral leakage is observed when comparing the spectrum of a signal with the spectrum of a short section of the same signal.
14. Extracting a section from a longer signal is equivalent to multiplying the signal by a rectangular windowing function.
15. Spectral leakage spreads each frequency sample of the original spectrum with a pattern defined by the DFT of the windowing function.
16. The effects of spectral leakage can be controlled by using a tapered windowing function instead of a rectangular one.
17. A rectangular window is preferable when spectral analysis aims to distinguish two spectral components with similar frequency and power.
18. A tapered window is preferable when leakage from a strong component may obscure a much weaker component at a different frequency.
19. The spectrum of a time-limited sampled signal is formally defined as the squared absolute value of its DFT, called periodogram.
20. In the averaged periodogram method, a long signal is split into shorter adjacent epochs, the spectrum of each epoch is computed, and these spectra are averaged bin by bin.
21. In Welch's method, the signal is split into shorter possibly overlapping epochs, each epoch is multiplied by a windowing function, and spectra are averaged bin by bin.
22. The Short-Time Fourier Transform is a simple method to estimate a spectrogram.
23. In a spectrogram, time is usually displayed on the horizontal axis, frequency on the vertical axis, and magnitude or power is color-coded.

2.27.4 Digital Filters

1. The purpose of a filter is to allow some spectral components of a signal to pass almost unaltered while blocking other spectral components.
2. Filters are categorized into four basic types: low-pass, high-pass, band-pass, and band-reject or band-stop.
3. The passband is the interval of frequencies in which the gain of the filter is close to 1.
4. The stopband is the interval of frequencies in which the gain is close to 0.
5. The transition band is the interval in which the gain has intermediate values between 0 and 1.
6. The roll-off of a filter is the slope of its frequency response in the transition band.
7. Roll-off is high when the transition band is narrow.
8. The cutoff frequency, or corner frequency, designates the limit of the passband.
9. In many filter designs, the gain at the cutoff frequency is approximately 0.71, corresponding to $-3\,\mathrm{dB}$.
10. The gain in the passband can decrease monotonically below 1 when the frequency approaches the cutoff frequency, or it may ripple above and below 1.
11. The frequency response of a filter in the stopband should preferably be evaluated with the gain expressed in dB.
12. Good features of a filter include fast roll-off, flat passband, and strong stopband attenuation.
13. Digital filters are categorized into FIR and IIR filters.
14. The output of FIR filters is a linear combination of input samples.
15. The output of IIR filters combines samples of the input and past samples of the output.
16. The Butterworth filter is a design method in the family of IIR filters.
17. The corner frequency of a high-pass filter is called low cutoff frequency.
18. The corner frequency of a low-pass filter is called high cutoff frequency.
19. Band-pass and band-stop filters have both a low cutoff frequency and a high cutoff frequency.
20. A notch filter is a type of band-stop filter that attenuates the input in a narrow band around the notch frequency.
21. The most common use of a notch filter is to remove the power-line artifact at 50 Hz and its harmonics, or 60 Hz in some countries.
22. An IIR filter is more efficient than a FIR filter, meaning that a FIR filter needs to be of higher order to achieve the same quality specifications.
23. A FIR filter can be designed to have linear phase, meaning it does not introduce time-domain distortions in the waveform of the output signal.
24. IIR filters cannot have linear phase.

3 Brain-Computer Interfaces

3.1 Definition of Brain-Computer Interface

A Brain-Computer Interface, BCI, is a system that measures brain activity and converts it into an artificial output. This output replaces, restores, enhances, supplements, or improves natural Central Nervous System output.

A key feature of a BCI is that it bypasses the normal neuromuscular pathways. Instead of using nerves and muscles to control the environment, the system extracts information directly from brain activity and translates it into commands.

$$\text{brain activity} \rightarrow \text{feature extraction} \rightarrow \text{translation algorithm} \rightarrow \text{command} \rightarrow \text{feedback.}$$

Author note.
A BCI is not simply a classifier applied to EEG. It is a closed-loop control system in which the user learns from feedback and modifies brain activity to improve control.

3.2 BCI Closed Loop

The BCI operates as a closed-loop pipeline:

1. the user generates an intention or cognitive state;
2. brain signals are acquired;
3. relevant features are extracted;
4. features are translated into a command;
5. the command acts on the environment;
6. the user receives feedback.

Feedback is essential because controlling the environment without muscles is unnatural. The brain needs feedback to learn how to modulate its signals effectively.

Author note.
The feedback loop is what allows BCI learning. Without feedback, the user cannot know whether the attempted mental strategy is producing the desired output.

3.3 Invasive and Non-Invasive BCIs

BCI signals can be acquired using invasive or non-invasive methods.

Invasive methods include:

- microelectrodes;
- single-unit activity;
- multi-unit activity;
- local field potentials;
- electrocorticography.

Non-invasive methods include:

- EEG;
- NIRS;

- fMRI.

EEG is one of the most widely used methods for non-invasive BCIs because it is relatively cheap, portable, and has good temporal resolution.

3.4 Assistive and Rehabilitative BCIs

Clinically, BCIs can be divided into two main paradigms:

- assistive BCIs;
- rehabilitative BCIs.

Assistive BCIs provide an alternative access channel for users with severe motor impairments. Their goal is not to heal the biological damage, but to allow communication and interaction with the environment.

Rehabilitative BCIs aim to drive neuroplasticity. Their goal is to improve biological function by linking motor intention to contingent sensory feedback.

> **Author note.**
> Assistive BCI replaces an output channel. Rehabilitative BCI tries to retrain the nervous system.

3.5 Assistive Technology and the Accessibility Gap

Traditional assistive devices, such as joysticks, touch screens, switches, head trackers, and eye trackers, require some reliable motor output.

In conditions such as ALS, locked-in syndrome, or severe stroke, even minimal muscular control may be unavailable. In these cases, a BCI can provide an alternative input channel that bypasses muscles.

The BCI does not need to replace the whole assistive software environment. In a modern modular architecture, the BCI simply outputs a discrete selection command, while standard assistive technology software manages communication, browsing, or domotic control.

3.6 P300 and the Oddball Paradigm

The P300 is an Event-Related Potential that appears as a positive voltage deflection approximately 300 ms after the presentation of a rare and meaningful target stimulus.

It is commonly elicited using the oddball paradigm, where rare target stimuli are presented among frequent non-target stimuli.

In a P300 BCI, the user attends to a target item. When that item flashes, a P300 response is generated. When non-target items flash, the P300 is absent or weaker.

3.7 P300 Speller

A classical assistive BCI application is the P300 Speller.

The interface consists of a matrix of letters and symbols. Rows and columns flash randomly. The user focuses attention on the desired character.

A P300 is generated when the row or column containing the desired character flashes. By detecting which row and column elicited the P300, the system infers the selected character.

Author note.

The P300 Speller is not based on motor imagery. It is based on attention to rare target stimuli and ERP detection.

3.8 P300 Feature Extraction and Classification

From a machine learning perspective, P300 decoding is a binary classification problem:

$$\text{target} \quad \text{vs.} \quad \text{non-target.}$$

Features are often constructed by taking EEG voltage values at specific post-stimulus latencies and channels. These values may be averaged over short temporal windows, for example around 50 ms.

The resulting feature vector is classified using linear methods such as SWLDA, Stepwise Linear Discriminant Analysis.

No exam / technical intuition.

Some detailed technical material about P300 feature extraction and SWLDA was marked as not discussed in class in the handout. It is still useful to understand why P300 BCIs work: the classifier is not reading "letters" directly, but detecting whether a target-related ERP was present after a flash.

3.9 Sensorimotor Rhythms and Motor Imagery BCIs

Sensorimotor rhythms, SMR, include rhythms in the mu and beta bands. They are generated over sensorimotor cortical areas.

During actual movement and kinesthetic motor imagery, SMR power decreases. This decrease is often described as Event-Related Desynchronization, ERD.

Motor imagery BCIs exploit the voluntary modulation of SMR power. For example, imagining movement of the left or right hand can modulate rhythms over contralateral sensorimotor areas.

Author note.

Kinesthetic motor imagery means imagining the feeling of performing a movement, not simply visualizing yourself from outside. This distinction matters because kinesthetic imagery engages motor networks more strongly.

3.10 Asynchronous SMR-BCIs

Unlike P300 BCIs, which detect discrete event-related responses after stimuli, asynchronous SMR-BCIs translate continuous, state-dependent modulation of brain rhythms into control commands.

For example, in a cursor-control task, the system continuously estimates PSD features in specific EEG bands and channels. These features are translated into a continuous output controlling cursor movement.

Author note.

P300 BCI is event-driven: the system asks, "was there a P300 after this flash?" SMR BCI is state-driven: the system asks, "what is the current sensorimotor rhythm state?"

3.11 PSD Estimation in Online BCI

In SMR-BCIs, the system must estimate spectral power in real time. This often uses PSD estimation methods such as Welch's method.

A key design problem is balancing:

- spectral resolution;
- update speed;
- feedback smoothness;
- classification accuracy.

The epoch used for spectral estimation may be longer than the update interval. Therefore, consecutive epochs overlap.

This system-level epoch overlap is different from the internal window overlap used by Welch's method.

No exam / technical intuition.

This technical distinction was marked as additional material, but it is important for understanding real-time BCI design. The BCI can update every 100 ms while using a longer 300 ms data epoch, because consecutive epochs overlap.

3.12 Rehabilitative BCI and Sensorimotor Loop

After stroke, the connection between motor intention and movement-related sensory feedback may be disrupted. A rehabilitative BCI can act as a synthetic bridge.

The patient performs kinesthetic motor imagery. If the BCI detects appropriate motor-related brain activity, it triggers an external device, such as Functional Electrical Stimulation, FES, or a robotic orthosis.

This generates sensory feedback contingent on motor intent, closing the sensorimotor loop.

$$\text{motor intent} \to \text{BCI detection} \to \text{external movement or stimulation} \to \text{sensory feedback.}$$

Author note.

The rehabilitative idea is Hebbian: pairing motor cortical activation with the correct sensory feedback may reinforce surviving pathways and promote useful plasticity.

3.13 Hybrid BCIs and Corticomuscular Coherence

Hybrid BCIs combine multiple signal sources. A hybrid system may use both EEG and EMG.

Corticomuscular Coherence, CMC, measures functional synchronization between cortical motor activity and peripheral muscle activation.

A CMC-based rehabilitative BCI can encourage correct coupling between motor cortex and target muscles, while discouraging compensatory strategies such as shoulder activation or mirror movements of the healthy limb.

No exam / technical intuition.

The handout marks the advanced CMC architecture as optional cutting-edge research. It is useful to understand the concept, but the detailed spectral mathematics behind coherence is beyond the core requirements.

3.14 Clinical Relevance and Brain Network Reconfiguration

BCI-supported rehabilitation can induce training-specific brain network reconfiguration. In the reported clinical framework, BCI training was associated with improved clinical outcomes and increased involvement of ipsilesional sensorimotor areas.

Interhemispheric connectivity changes can also be band-specific. In particular, increased coupling in beta bands is especially relevant because beta rhythms are directly related to motor function.

Author note.
This is an important translational point: a successful rehabilitative BCI should not only improve a score, but also produce measurable neurophysiological changes consistent with motor recovery.

3.15 Core Exam Takeaways

1. A BCI measures CNS activity and converts it into an artificial output.
2. A BCI bypasses normal neuromuscular pathways.
3. BCI operation is closed-loop and depends on feedback.
4. Assistive BCIs provide alternative access channels.
5. Rehabilitative BCIs aim to promote neuroplasticity and restore function.
6. P300 BCIs exploit rare target-related ERP responses.
7. The P300 Speller detects which row and column elicit a P300.
8. SMR-BCIs exploit voluntary modulation of mu and beta rhythms.
9. Motor imagery produces SMR desynchronization.
10. Online SMR-BCIs require real-time PSD estimation.
11. Rehabilitative BCIs close a broken sensorimotor loop using contingent feedback.
12. Hybrid CMC-BCIs combine EEG and EMG to encourage correct corticomuscular coupling.

3.16 True Statements

1. A BCI is a system that measures brain activity and converts it to artificial output that replaces, restores, enhances, supplements, or improves natural brain output and thereby changes the ongoing interactions between the brain and the environment.
2. The P300 ERP generated by attending a target stimulus is exploited to build virtual keyboards based on a BCI.
3. The amplitude of sensorimotor rhythms can be voluntarily modulated through motor imagery to build cursor control based on a BCI.
4. A Brain-Computer Interface completely bypasses normal neuromuscular pathways to measure Central Nervous System activity and convert it into artificial output.
5. The operation of a Brain-Computer Interface is a closed-loop pipeline that relies on environmental feedback to allow the user's brain to learn and consciously modify its signals.

6. In assistive applications, BCIs do not aim to heal biological damage, but rather serve as permanent, alternative access channels for users with severe motor impairments.
7. The P300 is an Event-Related Potential that manifests as a positive voltage deflection approximately 300 milliseconds after the presentation of a rare, meaningful target stimulus.
8. In a P300 BCI, single features are typically constructed by averaging EEG voltage over a short temporal window at specific post-stimulus latencies and spatial channels.
9. Linear classifiers like SWLDA are preferred in P300 BCIs because they generalize well even with small training sets and have interpretable weights.
10. In a modern modular Assistive Technology architecture, a P300 BCI solely outputs a discrete selection trigger, delegating complex application logic to mainstream commercial software.
11. Sensorimotor rhythms, SMR, include the mu rhythm and have spectral components in the alpha and beta bands.
12. Sensorimotor rhythms desynchronize and decrease in spectral power during both physical execution and kinesthetic imagination of movement.
13. Rather than detecting discrete events, asynchronous rehabilitative BCIs translate continuous, state-dependent modulation of SMR power into active motor intent commands.
14. To ensure a smooth and responsive cursor, the BCI system's update interval is set shorter than the minimum epoch duration required for an acceptable spectral estimate.
15. System-level overlap of consecutive data epochs is distinct from the algorithmic overlap of the shorter windows used internally by Welch's method.
16. Although Event-Related Desynchronization formally refers to a non-stationary phenomenon anchored to a discrete event, the term ERD is frequently used in BCI contexts to describe continuous, state-dependent drops in SMR power.
17. The primary goal of a rehabilitative BCI is to guide Central Nervous System plasticity by actively engaging damaged motor networks to restore impaired biological functions.
18. Kinesthetic motor imagery involves rehearsing the actual physiological sensation of moving a limb and successfully engages primary and supplementary motor areas.
19. A rehabilitative BCI acts as a synthetic bridge that closes the broken sensorimotor loop by delivering exogenous sensory feedback exactly when the patient's motor cortex generates endogenous motor intent.
20. Successful BCI-supported motor imagery training induces training-specific brain network reconfiguration, significantly increasing interhemispheric connectivity in beta bands.

4 EMG

4.1 Definition and Applications of EMG

Electromyography, EMG, is a method used to record the electrical activity generated by skeletal muscles.

EMG is used in several fields:

- neuromuscular physiology;
- diagnosis of neuromuscular disorders;
- posture analysis;
- ergonomics and occupational medicine;
- musculoskeletal physical therapy;
- gait analysis;
- sport science;
- EMG biofeedback;
- man-machine interfaces.

Author note.
EMG measures muscle electrical activity, not force directly. However, under controlled conditions, EMG amplitude is related to the level of muscle activation.

4.2 Muscle Fibers and Excitability

A muscle fiber is a single cell of muscular tissue. Its membrane is excitable, meaning that it can respond to depolarization by producing an action potential.

The action potential of a muscle fiber is called Muscle Fiber Action Potential, MFAP.

$$\text{depolarization} \rightarrow \text{MFAP}.$$

MFAPs propagate along the muscle fiber and are the elementary electrical sources contributing to the EMG signal.

4.3 Muscle Fiber Conduction Velocity

The speed at which an MFAP travels along the muscle fiber is called Muscle Fiber Conduction Velocity, MFCV.

Typical values are:

$$MFCV \approx 2 - 6\,\text{m/s}.$$

MFCV depends on several factors:

- ionic concentrations;
- intracellular pH;
- temperature;
- muscle fiber diameter and morphology;
- muscle fiber length;
- fiber type, such as slow or fast-twitch fibers;
- muscle fatigue;
- neuromuscular pathology.

Author note.
Fatigue often changes muscle fiber conduction velocity and spectral content of EMG. This is one reason why EMG frequency-domain features can be useful in fatigue analysis.

4.4 Motor Neuron and Motor Unit

A Motor Unit, MU, is the fundamental functional unit of the neuromuscular system. It consists of:

- one alpha motor neuron;
- all the muscle fibers innervated by that motor neuron.

When the motor neuron fires, all fibers belonging to that motor unit are activated.

$$\text{one alpha motor neuron} + \text{its muscle fibers} = \text{Motor Unit}.$$

4.5 Motor Unit Action Potential

The Motor Unit Action Potential, MUAP, is the electrical signal generated by a single motor unit.

It results from the spatial and temporal summation of the MFAPs generated by the fibers belonging to that motor unit:

$$MUAP = \sum MFAP.$$

Author note.
The surface EMG signal is not the activity of a single fiber. It is the superposition of many MUAPs from many active motor units, filtered by body tissues and electrode configuration.

4.6 Force Modulation: Recruitment and Rate Coding

The Central Nervous System modulates muscle force mainly through two mechanisms:

- recruitment;
- rate coding.

Recruitment is the activation of additional motor units. More active motor units generally produce greater force.

Rate coding is the increase in firing frequency of already active motor units.

Both mechanisms occur together, but their relative contribution changes with force level.

- at low force levels, recruitment is dominant;
- at high force levels and during rapid contractions, rate coding becomes more important.

Author note.
A simple intuition is: to increase force, the CNS can either recruit more motor units or make already active motor units fire faster.

4.7 EMG Amplitude and Muscle Activation

When more motor units are recruited and/or their firing rate increases, the EMG amplitude tends to increase.

However, the relationship between EMG amplitude and force is not always perfectly linear. It depends on:

- muscle geometry;
- electrode placement;
- contraction type;
- fatigue;
- cross-talk;
- tissue properties;
- motor unit synchronization.

Author note.
EMG amplitude is often used as an estimate of muscle activation, but it should not be interpreted as a direct force sensor without considering experimental conditions.

4.8 Volume Conduction

The electrical potentials generated by muscle fibers propagate through biological tissues before reaching the recording electrodes.

These tissues include:

- muscle tissue;
- fat;
- skin;
- connective tissue.

Together, they act as a volume conductor.

Biological tissues spatially and temporally low-pass filter the EMG signal. This means that high spatial and high temporal frequency components are attenuated before reaching the electrodes.

Author note.
Volume conduction explains why surface EMG is a blurred version of the underlying muscle electrical activity. The electrode does not see the source directly; it sees the source after propagation through tissues.

4.9 Tripole Model of the MFAP

The propagating MFAP can be modeled as a tripole, with:

- a central negative phase;
- two leading and trailing positive phases.

As the tripole propagates along the muscle fiber, the recorded potential changes depending on the relative position of the source and the electrode.

End-fiber effects occur when the action potential reaches the end of the muscle fiber. These components may decay with distance differently from the propagating components.

4.10 Electrode Types and Placement

EMG can be recorded using different electrode types:

- surface electrodes;
- needle or intramuscular electrodes.

Surface EMG is non-invasive and suitable for global muscle activation analysis. Intramuscular EMG is more selective but invasive.

Electrode placement strongly affects the recorded signal. Important factors include:

- electrode position relative to the muscle belly;
- distance from the innervation zone;
- orientation with respect to muscle fibers;
- inter-electrode distance;
- skin preparation;
- cross-talk from nearby muscles.

Author note.
Bad electrode placement can change EMG amplitude and frequency content more than the physiological effect you are trying to measure. Placement is therefore not a minor detail.

4.11 Selectivity and Cross-Talk

Selectivity is the ability of the EMG recording to capture activity from the target muscle while rejecting activity from other muscles.

Cross-talk occurs when the EMG signal includes activity from nearby muscles.

Cross-talk is influenced by:

- electrode size;
- inter-electrode distance;
- depth of the target muscle;
- thickness of subcutaneous tissue;
- anatomical proximity of muscles.

Author note.
Surface EMG is convenient but not perfectly selective. If two nearby muscles activate together, the electrode may pick up both.

4.12 EMG Signal Processing

Typical EMG processing includes:

1. acquisition and amplification;
2. filtering;
3. artifact inspection;
4. rectification;
5. smoothing or envelope extraction;
6. feature extraction.

A common preprocessing pipeline is:

$$\text{raw EMG} \to \text{band-pass filtering} \to \text{rectification} \to \text{low-pass smoothing} \to \text{envelope}.$$

Rectification transforms negative values into positive values:

$$x_{\text{rect}}(t) = |x(t)|.$$

The envelope represents the slow modulation of EMG amplitude over time.

Author note.
Rectification is necessary because raw EMG oscillates around zero. Without rectification, positive and negative phases would cancel during averaging or smoothing.

4.13 EMG Features

EMG features can be extracted in several domains:

- amplitude domain;
- time domain;
- frequency domain.

Common amplitude and time-domain features include:

- RMS;
- ARV or MAV, mean absolute value;
- integrated EMG;
- zero crossings;
- waveform length.

Frequency-domain features include:

- power spectral density;
- mean frequency;
- median frequency.

Author note.
RMS and ARV are often used as proxies for muscle activation level. Mean and median frequency are often used when studying fatigue or changes in conduction velocity.

4.14 EMG in Man-Machine Interfaces

EMG can be used as an input signal for man-machine interfaces. For example, muscle activation patterns can be translated into commands for:

- prosthetic control;
- orthotic control;
- rehabilitation devices;
- gesture recognition systems;
- human-computer interaction.

Compared with EEG, EMG usually has larger amplitude and higher signal-to-noise ratio, but it depends on the availability of residual muscular activity.

4.15 Core Exam Takeaways

1. EMG records electrical activity generated by skeletal muscles.
2. A muscle fiber is an excitable cell.
3. Depolarization of a muscle fiber produces an MFAP.
4. MFCV is typically in the range 2–6 m/s.
5. A motor unit consists of one alpha motor neuron and all the muscle fibers it innervates.
6. A MUAP is the spatial and temporal sum of MFAPs from one motor unit.
7. Muscle force is modulated by recruitment and rate coding.
8. Recruitment dominates at low force levels.
9. Rate coding becomes more important at high forces and during rapid contractions.
10. Surface EMG is affected by volume conduction.
11. Biological tissues act as spatial and temporal low-pass filters.
12. The propagating MFAP can be modeled as a tripole.
13. Electrode placement strongly affects the recorded EMG.
14. Cross-talk is contamination from nearby muscles.
15. EMG processing often includes filtering, rectification, smoothing, and feature extraction.

4.16 True Statements

1. A muscle fiber is a single cell of the muscular tissue.
2. The membrane of the muscle fiber is an excitable tissue that produces a Muscle Fiber Action Potential, MFAP, upon depolarization.
3. The speed at which an MFAP travels along the muscle fiber is the Muscle Fiber Conduction Velocity, MFCV.
4. MFCV is typically in the range of 2–6 m/s.
5. MFCV is influenced by factors such as muscle temperature, fiber diameter, and fatigue.
6. A Motor Unit, MU, is the fundamental functional unit of the neuromuscular system.
7. A Motor Unit consists of a single alpha motor neuron and all the muscle fibers it innervates.
8. The Motor Unit Action Potential, MUAP, is the electrical signal generated by a single motor unit.
9. The MUAP represents the spatial and temporal sum of all constituent MFAPs.
10. The Central Nervous System modulates muscle force through two primary mechanisms: recruitment and rate coding.
11. Recruitment is the process of activating additional motor units to increase the force of contraction.

12. Rate coding refers to increasing the firing frequency, in impulses per second, of already active motor units.
13. At low force levels, recruitment is the dominant mechanism for force modulation.
14. At high force levels and during rapid contractions, rate coding becomes a more significant contributor to force generation.
15. The combination of recruitment and rate coding differs between increasing and decreasing force.
16. More recruited motor units and greater rate coding generally mean that EMG amplitude increases.
17. The tissues between the muscle fibers and the recording electrodes, such as fat and skin, act as a volume conductor.
18. The volume conductor spatially and temporally low-pass filters the EMG signal.
19. The propagating MFAP can be modeled as a tripole, with a central negative phase and two leading or trailing positive phases.
20. End-fiber effects fall off with distance less than the other components of the MUAP.
21. Temporal high-pass and low-pass components of the MUAP are linked to its spatial components because the source travels with a specific conduction velocity.

Part II
Prof. Astolfi

5 Introduction to Neuroengineering

5.1 Why a Neuroengineering Course?

Neuroengineering studies the nervous system using concepts, tools, and models from engineering. The human brain can be seen as a complex learning system able to process a very large flow of information and transform it into actions at a temporal scale of milliseconds.

The brain has inspired many engineering solutions, including artificial intelligence, robotics, control systems, neural prostheses, and brain-computer interfaces. At the same time, engineering provides tools to study and model brain activity, interpret neural signals, and develop clinical or assistive technologies.

Author note.
The key idea of this part of the course is that the relationship between neuroscience and engineering is bidirectional. Neuroscience inspires engineering models, such as artificial neural networks, while engineering provides methods to understand, measure, and interact with the nervous system.

5.2 Physiologically Inspired Devices

One historical example of a physiologically inspired device is the perceptron, invented in 1958 by Frank Rosenblatt. The perceptron was designed according to biological principles and was able to learn, making it one of the early precursors of artificial intelligence.

Another example is social robotics, which studies robots able to interact and communicate with humans, other robots, and the environment according to social rules and roles.

> **Author note.**
> The perceptron is important not because it is biologically realistic in every detail, but because it shows how a simplified idea inspired by neurons can become an engineering model for learning and classification.

5.3 Control Theory, Robotics, AI, and Neuroscience

Control theory is useful in neuroscience because the brain can be interpreted as a complex control system. It receives inputs, processes information, and produces outputs such as motor actions, cognitive responses, and behavioral decisions.

In this view:

- sensory information acts as input;
- the brain and central nervous system act as controller;
- muscles and body dynamics act as plant;
- motor commands act as output.

Robotics and neural prostheses use control algorithms to interpret neural signals and produce appropriate actions. Brain-computer interfaces, BCIs, use control engineering principles to translate neural activity into commands for external devices.

Artificial intelligence also plays a central role in neuroscience. It can help analyze large and complex neural datasets, identify patterns in brain data, improve brain imaging interpretation, support hypothesis testing, and contribute to personalized medicine and neuroprosthetics.

> **Author note.**
> For an AI and Robotics student, this is the conceptual bridge: the nervous system is not only a biological object, but also an information-processing and control system. This makes it natural to study it using signal processing, control, machine learning, and network models.

5.4 Main Topics of Astolfi's Module

The module includes the following main topics:

1. physiology of the neuron: generation, integration, and propagation of neural electrical signals;
2. principles of neuroanatomy and brain organization;
3. mechanisms of generation of neural electrical correlates;
4. neural encoding and decoding;
5. basic definitions of network neuroscience, such as synchronicity, causality, and influence;
6. brain functional networks at different scales;
7. graph theoretical analysis of brain networks;
8. multi-subject systems;
9. applications to clinical and physiological problems.

6 The Neural Cell

6.1 Main Functions of the Neuron

A neuron has three main functions:

1. collection of information from multiple sources, such as other neurons or receptors;
2. integration of incoming information in space and time, producing a binary decision;
3. generation and propagation of a bit of information toward target cells, such as other neurons or muscle cells.

The specialized structure of the neuron supports these functions. Dendrites collect inputs, the soma and axon hillock integrate information, and the axon propagates the output signal toward synaptic terminals.

Author note.
A compact way to remember the neuron is: dendrites collect, the axon hillock decides, the axon transmits.

6.2 The Neuronal Membrane

The neuronal membrane is the main morphologically specialized structure that allows neurons to produce electrical signals. It is a phospholipid bilayer with hydrophilic heads and hydrophobic tails. The membrane is selectively permeable to ions and contains membrane proteins such as ion channels and receptors.

The main ion families involved in neuronal electrical behavior are:

- sodium, Na^+;
- potassium, K^+;
- chloride, Cl^-;
- calcium, Ca^{2+}.

Ion channels allow passive transport driven by electrochemical forces. Ion pumps allow active transport against electrochemical gradients and require energy expenditure by the cell.

Author note.
The membrane is not just a passive boundary. It is an active computational interface: by controlling which ions can cross and when, the neuron changes its electrical state.

6.3 Electrochemical Equilibrium

Ion movement across the neuronal membrane is driven by two types of forces:

- diffusional forces, due to concentration gradients;
- electrical forces, due to voltage differences across the membrane.

The balance between these two forces determines the electrochemical equilibrium. For a given ion family, the electrochemical potential difference can be written as:

$$\Delta\mu = RT \ln\left(\frac{[X]_A}{[X]_B}\right) + zF(E_A - E_B),$$

where:

- $[X]_A$ and $[X]_B$ are the ion concentrations on the two sides of the membrane;
- R is the universal gas constant;
- T is the temperature in Kelvin;
- F is the Faraday constant;
- z is the ion valence;
- $E_A - E_B$ is the membrane potential difference.

At electrochemical equilibrium, $\Delta\mu = 0$.

6.4 Nernst Equation

The Nernst equation gives the equilibrium potential for a specific ion family j:

$$E_j = \frac{RT}{Fz_j} \ln\left(\frac{[extra]_j}{[intra]_j}\right),$$

where:

- $[extra]_j$ is the extracellular concentration of ion j;
- $[intra]_j$ is the intracellular concentration of ion j;
- z_j is the valence of ion j.

The equilibrium potential E_j is the membrane voltage at which diffusional and electrical forces balance for ion j. If the membrane is freely permeable to that ion, the net current tends to move the membrane potential toward E_j.

Typical equilibrium potentials are:

Ion	**Typical equilibrium potential**
K^+	$E_K \approx -90\,\text{mV}$
Na^+	$E_{Na} \approx +50\,\text{mV}$
Ca^{2+}	$E_{Ca} \approx +150\,\text{mV}$

Author note.
The Nernst equation does not describe the full membrane potential by itself. It gives the equilibrium voltage for one ion species. The actual resting membrane potential depends on the combined effect of several ions, their permeability, and ion pumps.

6.5 Resting Membrane Potential

The membrane potential is the difference in electrical potential between the inside of the neuron and the extracellular fluid. It is caused by different ion concentrations on the two sides of the membrane.

At rest, the neuronal membrane potential is approximately:

$$V_{\text{rest}} \approx -70\,\text{mV}.$$

This means that the inside of the neuron is more negative than the outside. The membrane is therefore said to be polarized.

The resting membrane potential depends on:

1. diffusional forces;
2. electrical forces;
3. ion pumps.

The most important ion pump is the sodium-potassium pump, which moves ions against their electrochemical gradients using energy.

6.6 Depolarization, Repolarization, and Hyperpolarization

A membrane depolarization occurs when the membrane potential becomes less negative or even positive. This can happen when positive ions enter the cell or negative ions leave the cell.

A membrane hyperpolarization occurs when the membrane potential becomes more negative than the resting value. This can happen when positive ions leave the cell or negative ions enter the cell.

Repolarization is the return of the membrane potential toward the resting value after a depolarization.

Author note.

For exam purposes, it is useful to associate the direction of voltage change with the functional effect: depolarization brings the neuron closer to firing an action potential; hyperpolarization usually makes firing less likely.

6.7 Dendritic Tree and Synapses

The dendritic tree collects information from other neural cells or receptors through synaptic connections. A single neuron may receive from thousands up to hundreds of thousands of inputs.

There are two main types of synapses:

- electrical synapses;
- chemical synapses.

Electrical synapses allow direct electrical transmission between cells.

Chemical synapses use neurotransmitters. The presynaptic neuron releases a chemical signal into the synaptic cleft; this neurotransmitter binds to receptors on the postsynaptic membrane and causes the opening of specific ion-gated channels.

6.8 Excitatory and Inhibitory Synapses

A postsynaptic potential can be excitatory or inhibitory.

An excitatory postsynaptic potential, EPSP, causes depolarization:

$$\text{EPSP} \to \text{depolarization.}$$

An inhibitory postsynaptic potential, IPSP, causes hyperpolarization:

$$\text{IPSP} \to \text{hyperpolarization.}$$

Excitatory and inhibitory are defined with respect to the probability of generating a neuronal response. An EPSP makes the neuron more likely to fire an action potential, while an IPSP makes it less likely.

Author note.

Excitatory does not mean “good” and inhibitory does not mean “bad”. They simply describe whether the postsynaptic effect increases or decreases the probability of firing.

6.9 Temporal and Spatial Summation

The neuron integrates incoming information by summing EPSPs and IPSPs with their sign.

There are two main forms of summation:

- temporal summation: multiple postsynaptic potentials arrive close together in time from the same or similar input;
- spatial summation: postsynaptic potentials arrive from different synaptic locations on the neuron.

Temporal and spatial summation can occur simultaneously. The final membrane depolarization or hyperpolarization is propagated toward the axon hillock. If the resulting membrane potential reaches threshold, the neuron fires an action potential.

6.10 Propagation of Postsynaptic Potentials

Postsynaptic potentials propagate passively along the membrane. A local depolarization or hyperpolarization causes intra- and extracellular ionic currents that spread to adjacent membrane regions.

However, this propagation is:

- inefficient;
- non-directional;
- decreasing with distance;
- prevalent in dendrites.

Author note.
Postsynaptic potentials are analog and graded: their amplitude can vary, and they decay with distance. This is different from the action potential, which is all-or-none and actively propagated.

6.11 Action Potential

The action potential is a rapid variation of the membrane potential that appears in neural, muscular, and cardiac cells. In neurons, it represents the binary output decision of the cell.

It is an all-or-none process:

- if the stimulus does not reach threshold, no action potential occurs;
- if threshold is reached, the action potential has the same shape, duration, and amplitude, independently of the stimulus amplitude.

Typical properties of an action potential are:

- duration: approximately 2 ms;
- amplitude: approximately 100 mV;
- spike-like waveform.

The information is not encoded in the shape of the action potential, but in the temporal distance between spikes, that is, in the spike train.

6.12 Voltage-Gated Sodium and Potassium Channels

The action potential depends mainly on voltage-gated Na^+ and K^+ channels.

For Na^+ channels:

- at resting potential, around $-70\,\text{mV}$, the channel is closed but can be opened;
- at the opening threshold, around $-60\,\text{mV}$, the channel opens and Na^+ enters the cell;
- near $+30\,\text{mV}$, the channel becomes inactivated and cannot be opened;
- when the membrane returns toward rest, the channel becomes closed again.

For K^+ channels:

- at rest, they are closed;
- near the positive peak of the action potential, they open;
- K^+ flows out of the cell, contributing to repolarization;
- they close when the membrane becomes sufficiently negative again.

6.13 Phases of the Action Potential

The sequence of events is:

1. depolarization reaches the Na^+ opening threshold, around $-60\,\text{mV}$;
2. fast depolarization occurs because Na^+ flows into the cell;
3. Na^+ channels inactivate and K^+ channels open near the positive peak;
4. repolarization occurs because K^+ flows out of the cell;
5. undershoot or hyperpolarization occurs until K^+ channels close;
6. the membrane returns to resting potential, with the help of ion pumps.

6.14 Absolute and Relative Refractory Periods

The absolute refractory period is caused by the inactivation of voltage-gated Na^+ channels. During this period, no new action potential can be generated under any circumstances.

The relative refractory period is mainly related to K^+ channel dynamics and hyperpolarization. During this period, a new action potential can occur, but it requires a stronger depolarization.

Author note.

The refractory periods are essential because they limit the firing rate of the neuron and help make action potential propagation unidirectional.

6.15 Propagation of the Action Potential

There are two main mechanisms of action potential propagation:

- continuous conduction;
- saltatory conduction.

In continuous conduction, the action potential is regenerated along adjacent membrane regions. The absolute refractory period prevents the signal from propagating backward, producing unidirectional propagation. The propagation speed is relatively low, around $1\,\text{m/s}$, and is limited by the activation of Na^+ and K^+ channels.

In saltatory conduction, the axon is myelinated and the action potential is generated only at the nodes of Ranvier. Along myelinated segments, propagation is passive. The signal appears to “jump” from one node to the next, making saltatory conduction much faster than continuous conduction.

Author note.
The myelin sheath acts like electrical insulation. It prevents current leakage along the covered axon segments and allows the signal to travel quickly between nodes of Ranvier.

7 Principles of Neuroanatomy and Brain Organization

7.1 Studying the Human Brain

The brain can be studied through several approaches, including lesion studies, animal studies, neuroimaging, neuromodulation, behavioral data, and clinical data.

A central idea is that the brain predicts the future on the basis of the past, helping the individual survive and adapt.

Brain functioning occurs at different scales:

- temporal scale: milliseconds;
- spatial scale: from nanometers and micrometers to millimeters and centimeters.

Brain evolution occurs both collectively, across species and generations, and individually, because the brain remains plastic during the lifespan.

7.2 Grey Matter, White Matter, and Cortex

Grey matter mainly contains neuronal cell bodies and dendrites. White matter mainly contains myelinated axons, which form fiber pathways connecting different brain regions.

White matter tractography is an in vivo procedure based on diffusion tensor imaging, DTI, used to reconstruct white matter pathways.

The corpus callosum is a major white matter structure connecting the two hemispheres and allowing information exchange between them.

The brain cortex is highly folded to increase cortical surface within a limited volume. The folds are called gyri, while the grooves are called sulci. A large part of the cortical surface lies inside the sulci.

7.3 Cortical Organization

The cerebral cortex is organized into six vertical layers, which are strongly interconnected. Cortical neurons are also organized in columns, approximately 0.5 mm in diameter, perpendicular to the cortical surface.

Neurons within the same column tend to show similar responses to the same stimulus.

Author note.
A useful hierarchy is: neurons form layers and columns; columns form areas; areas interact in circuits and networks.

7.4 Types of Neural Connections

Neural connections can be classified as:

- feedforward or bottom-up connections: from early processing stages to later ones;
- lateral connections: between regions at the same processing level;
- feedback or top-down connections: from advanced processing stages back to earlier ones.

Most connections are lateral. All these connection types can be excitatory or inhibitory.

7.5 Anatomical Orientation Terms

Some common orientation terms are:

Term	**Meaning**
Anterior	In front of
Posterior	Behind
Superior	Above
Inferior	Below
Rostral	Toward the front of the brain
Caudal	Toward the back of the brain
Ventral	Toward the belly
Dorsal	Toward the back
Medial	Toward the midline
Lateral	Away from the midline
Ipsilateral	On the same side
Contralateral	On the opposite side

7.6 Brain Lobes and Main Functions

Lobe	**Main functions**
Frontal	Executive functions, reasoning, decision-making, expressive language, high-level cognition, orientation, planning and execution of movement.
Parietal	Primary and secondary somatosensory cortex, spatial navigation, touch, pressure, temperature, and pain.
Occipital	Primary visual cortex and visual information processing.
Temporal	Auditory cortex, receptive language, memory formation, and emotion through structures such as the hippocampus.

7.7 Brodmann Areas

Brodmann areas were first described by Korbinian Brodmann in 1909. They are based on cytoarchitecture, meaning the type and organization of neural cells. Later, they were associated with specific functions.

A single function may involve multiple areas. For example, Brodmann areas 3, 1, and 2 all contribute to the sensory cortex.

7.8 Feature Maps

Some brain regions are organized as feature maps.

In the visual cortex, there is retinotopic organization: the visual image from the retina is transformed and mapped onto the primary visual cortex, V1. Specialized neuronal populations process features such as shape, motion, and contrast.

In the motor and somatosensory cortex, there is somatotopic organization. The motor control area is located in the precentral gyrus of the frontal lobe, corresponding to Brodmann area 4. Somatosensory areas are located in the postcentral gyrus of the parietal lobe, corresponding mainly to Brodmann areas 3, 1, and 2.

The Penfield homunculus represents the somatotopic organization of motor and somatosensory cortices.

7.9 Subcortical Areas

Main subcortical areas include:

- thalamus;
- brainstem;
- basal ganglia;
- cerebellum.

The thalamus controls traffic from the periphery to the cortex, especially sensory afferent information. Subcortical areas are more difficult to reach and measure, especially using non-invasive methods.

7.10 Brain Organization at Different Levels

Brain organization can be studied at multiple spatial scales:

- cortical layers;
- cortical columns;
- Brodmann areas and cortical regions;
- brain circuits and networks.

These levels span different spatial scales, from micrometers to centimeters.

7.11 Brain Plasticity

Brain plasticity is the capacity of the nervous system to change its structure and function. Plasticity is maximal during the critical period, approximately from birth to two years of age, when brain reorganization can produce irreversible changes.

However, plasticity is also present in adults and elderly people. Evidence of adult plasticity includes:

- learning;
- memory;
- effects of sensory deprivation;
- reorganization after brain lesions;
- effectiveness of neurorehabilitation.

A key principle is Hebbian learning: cells or systems of cells that are repeatedly active together tend to become associated, so that activity in one facilitates activity in the other.

Author note.
The informal version of Hebbian learning is: neurons that fire together wire together. This does not describe all plasticity mechanisms, but it is a very useful intuition.

7.12 Main Mechanisms of Brain Plasticity

The main neuronal mechanisms behind plasticity are:

1. synaptic plasticity;
2. axonal sprouting or pruning;
3. axonal regeneration;
4. unmasking of latent synaptic connections.

Synaptic plasticity modifies the strength of existing synapses. These changes may occur over hours or weeks and can involve variations in neurotransmitter release or receptor availability.

Axonal sprouting creates new collateral branches to fill vacant synaptic sites, for example after the loss of an axon. Pruning removes unnecessary connections.

Axonal regeneration is typical of early life stages and is less common in adults.

Unmasking of latent synaptic connections refers to the use of redundant synapses that already exist but are not normally active until they are needed, for example after a lesion.

Author note.
Plasticity is the biological basis for learning, adaptation, and neurorehabilitation. However, it is not always positive: maladaptive plasticity can also contribute to pathological conditions.

8 Neural Encoding and Poisson Spike Generation

8.1 Neural Encoding and Neural Decoding

Neural encoding and decoding study the relationship between a stimulus feature and the action potentials generated by a neuron in response to that stimulus.

A stimulus may be external, such as light intensity, sound intensity, orientation, or movement direction, or internal, such as the direction of a planned movement.

In neural encoding, the direction of the problem is:

$$\text{stimulus} \longrightarrow \text{neural response}.$$

The aim is to describe how neurons react to different stimuli and to predict their response to a new stimulus:

$$\text{neural response} = f(\text{stimulus}).$$

In neural decoding, the direction is reversed:

$$\text{neural response} \longrightarrow \text{stimulus}.$$

The aim is to infer which stimulus, or which stimulus property, produced a given spike train:

$$\text{stimulus} = f^{-1}(\text{neural response}).$$

Author note.
Encoding asks: “given the stimulus, what does the neuron do?” Decoding asks: “given what the neuron did, what stimulus was likely present?” This distinction is very important for BCIs, because a BCI usually receives neural activity and must infer the user’s intention.

8.2 Rate-Coding Hypothesis

The rate-coding hypothesis states that the frequency of action potentials, or firing rate, is a fundamental mechanism used by neurons to encode information.

A classical example is the relation between pressure applied to a cat’s footpad and the number of action potentials generated by cutaneous nerve fibers. Higher pressure produces a higher number of spikes.

Author note.

Rate coding does not mean that spike timing is irrelevant in all contexts. It means that, in this simplified model, the relevant information is summarized by the number of spikes per unit time, rather than by the exact timing of every spike.

8.3 Spike Train as a Neural Response Function

A spike train can be represented mathematically as a sum of Dirac delta functions. If a neuron emits n spikes during a trial of duration T, and the i-th spike occurs at time t_i, then the neural response function is:

$$\rho(t) = \sum_{i=1}^{n} \delta(t - t_i), \qquad 0 \leq t_i \leq T.$$

Each spike is represented as an ideal instantaneous event.

8.4 Spike-Count Firing Rate

The spike-count firing rate is the time average of the neural response function over the duration of the trial:

$$r = \frac{n}{T} = \frac{1}{T} \int_0^T \rho(\tau)\, d\tau.$$

Since:

$$\int_0^T \rho(\tau)\, d\tau = n,$$

the firing rate is simply the number of spikes divided by the trial duration. Its unit is Hz. For example, if a neuron fires 6 spikes in 1 second:

$$r = \frac{6}{1} = 6\,\mathrm{Hz}.$$

Author note.

The spike-count firing rate is meaningful when the stimulus is stationary during the trial. If the stimulus changes over time, a single number $r = n/T$ may be too crude, and a time-resolved firing rate is needed.

8.5 Variability Across Trials

Even if the same stimulus is presented multiple times, neural responses may vary across trials. For example, the same stimulus may produce firing rates such as:

$$r_1 = 6\,\mathrm{Hz}, \qquad r_2 = 4\,\mathrm{Hz}, \qquad r_3 = 5\,\mathrm{Hz}.$$

For this reason, neural responses are often described probabilistically, using an average firing rate across trials:

$$\langle r \rangle.$$

Author note.

This is one of the reasons neuroscience often uses probability. Neurons are not deterministic machines at the level of single trials. The same stimulus can produce slightly different spike trains.

8.6 Tuning Curves

A stimulus usually has many properties, such as orientation, contrast, shape, movement direction, or disparity. In a tuning curve experiment, one property s is selected and varied, while the neural response is measured.

The tuning curve describes how the average firing rate changes as a function of the stimulus property:

$$r = f(s).$$

The general experimental procedure is:

1. choose a stimulus property s;
2. present several values of s;
3. repeat each condition multiple times;
4. compute the average firing rate for each value of s;
5. plot $\langle r \rangle$ as a function of s.

8.7 Example: Orientation Tuning in Visual Cortex

A neuron in the primary visual cortex may respond selectively to the orientation of a visual bar. In this case, the stimulus property is the orientation angle s.

A possible tuning curve is Gaussian-shaped:

$$f(s) = r_{\max} \exp\left[-\frac{1}{2}\left(\frac{s - s_{\max}}{\sigma_f}\right)^2\right],$$

where:

- $s_{\max}$ is the angle evoking the maximum response;
- $r_{\max}$ is the maximum firing rate;
- σ_f controls the width of the tuning curve.

A narrow tuning curve means that the neuron is highly selective for a specific stimulus value. A broad tuning curve means that the neuron responds to a wider range of values.

8.8 Example: Direction Tuning in Motor Cortex

A neuron in the primary motor cortex may be tuned to movement direction. A common model is:

$$f(s) = r_0 + (r_{\max} - r_0) \cos(s - s_{\max}),$$

where:

- s is the movement direction;
- $s_{\max}$ is the preferred movement direction;
- $r_{\max}$ is the maximum firing rate;
- r_0 is an offset used to avoid negative firing rates.

Author note.
In motor cortex, a neuron usually does not encode a full movement by itself. Instead, many neurons with different preferred directions contribute together. This is the idea behind population coding.

8.9 Example: Disparity Tuning

Some visual neurons respond to binocular retinal disparity, which is the difference in the retinal position of an image between the two eyes. A possible tuning curve is sigmoidal:

$$f(s) = \frac{r_{\max}}{1 + \exp\left(\frac{s_{1/2} - s}{\Delta s}\right)},$$

where:

- s is binocular retinal disparity;
- $s_{1/2}$ is the disparity producing half of the maximum response;
- Δs controls how quickly the firing rate increases with s.

8.10 Poisson Spike Generator

A Poisson spike generator is a probabilistic model used to simulate spike trains. The main assumption is that spikes occur independently, with a probability determined by the firing rate.

If the firing rate is constant, the number of spikes in a time interval follows a Poisson distribution:

$$P(n) = \frac{(rT)^n}{n!} e^{-rT},$$

where:

- n is the number of spikes;
- r is the firing rate;
- T is the duration of the observation window.

The average number of spikes is:

$$\langle n \rangle = rT.$$

Author note.
The Poisson model is useful because it gives a simple bridge between firing rate and spike variability. If the firing rate increases, higher spike counts become more probable. However, it is only an approximation of real neuronal behavior.

8.11 Inter-Spike Intervals

In a Poisson process, the inter-spike interval, ISI, follows an exponential distribution. Long inter-spike intervals have a probability that decreases exponentially with their duration.

$$P(\text{ISI} > t) = e^{-rt}.$$

This means that the probability of waiting a long time before the next spike decreases as the firing rate increases.

8.12 Limitations of the Poisson Generator

The basic Poisson generator has important limitations:

- it assumes that each spike is independent from the others;
- it does not account for the absolute and relative refractory periods;
- it allows arbitrarily short inter-spike intervals;
- it may not reproduce the ISI distribution observed in real neurons.

A more realistic spike generator can include constraints on the minimum inter-spike interval, reflecting the refractory period of real neurons.

Author note.
The main missing biological detail in a simple Poisson generator is the refractory period. A real neuron cannot fire two spikes arbitrarily close in time, because after an action potential sodium channels are temporarily inactivated.

9 Neural Decoding

9.1 Meaning and Need for Neural Decoding

Neural decoding is the estimation of the stimulus or mental state that produced a given neuronal output. Formally:

$$\text{stimulus} = f^{-1}(\text{neural response}).$$

In probabilistic terms, encoding and decoding can be written as conditional probabilities:

$$\text{Encoding: } P(r \mid s),$$

$$\text{Decoding: } P(s \mid r).$$

Encoding asks for the probability of a response r given a stimulus property s. Decoding asks for the probability of a stimulus property s given that the neural response was r.

Author note.
Decoding is essential for brain-computer interfaces. The external device does not know the stimulus, intention, or movement plan directly. It only observes neural activity and must infer what it means.

9.2 Visual Discrimination Example

In the visual discrimination task described in the slides, a subject is trained to recognize the direction of motion of dots on a screen.

There are two possible movement directions:

- a direction corresponding to maximum neural response, denoted as $+$;
- a direction corresponding to minimum neural response, denoted as $-$.

A second factor is coherence, defined as the percentage of dots moving in the same direction. Coherence controls the amount of noise in the stimulus:

- 100% coherence: all dots move coherently;
- 50% coherence: only part of the motion information is coherent;
- 0% coherence: dots move randomly.

Author note.
Coherence is a way to manipulate task difficulty. At high coherence, the stimulus is easy to classify. At low coherence, neural responses to the two directions become more overlapping, so decoding becomes harder.

9.3 Firing Rate Distributions

For each stimulus condition, the firing rate is not a single deterministic value but a distribution. If the two distributions corresponding to two stimuli are well separated, decoding is easy. If they overlap strongly, decoding is difficult.

The separation between firing rate distributions determines how well a decoder can infer the stimulus.

9.4 Discriminability Index

The discriminability index d' measures how well two response distributions can be separated. A common definition is:

$$d' = \frac{\mu_+ - \mu_-}{\sqrt{\frac{1}{2}(\sigma_+^2 + \sigma_-^2)}},$$

where:

- μ_+ and μ_- are the mean firing rates for the two stimulus classes;
- σ_+^2 and σ_-^2 are their variances.

Large d' means high discriminability. Small d' means strong overlap and poor discriminability.

Author note.
d' increases when the means of the two classes move far apart or when the variability of the responses decreases. A neuron with noisy responses may be hard to decode even if its average response differs between conditions.

9.5 Threshold-Based Classification

A simple decoding strategy is to classify the stimulus using a threshold z on the firing rate.

For example:

$$r > z \Rightarrow \text{class } +,$$

$$r \leq z \Rightarrow \text{class } -.$$

Changing z changes the behavior of the classifier. A low threshold may detect many positive cases but also produce many false positives. A high threshold may reduce false positives but increase false negatives.

9.6 Confusion Matrix: TP, TN, FP, FN

The classification result can be summarized using:

- True Positive, TP: the stimulus is positive and the classifier says positive;
- True Negative, TN: the stimulus is negative and the classifier says negative;
- False Positive, FP: the stimulus is negative but the classifier says positive;
- False Negative, FN: the stimulus is positive but the classifier says negative.

From these values, one can define:

$$\text{True Positive Rate} = \frac{TP}{TP + FN},$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN}.$$

9.7 ROC Curve

The Receiver Operating Characteristic, ROC, curve describes classifier performance as the threshold z changes.

The ROC curve plots:

$$\text{False Positive Rate} \quad \text{versus} \quad \text{True Positive Rate.}$$

A ROC curve close to the diagonal corresponds to random classification. A curve close to the upper-left corner corresponds to high accuracy.

9.8 Area Under the Curve

The Area Under the Curve, AUC, summarizes the ROC curve in a single number.

$$0.5 \leq AUC \leq 1.$$

- $AUC = 0.5$: chance-level classification;
- $AUC = 1$: perfect classification.

Author note.
AUC is useful because it evaluates the classifier across all possible thresholds. Accuracy at a single threshold may be misleading if the threshold was chosen badly.

9.9 Subject and Classifier Performance

If neural decoding performance is close to behavioral performance, one may hypothesize that the neural population under study is actually involved in the perceptual or decision process used by the subject.

Author note.
This does not prove causality by itself. Similarity between neural decoding and behavior suggests relevance, but proving that a neural population causes the behavior requires stronger evidence, such as perturbation or intervention.

10 Brain Networks I: Correlation and Coherence

10.1 Why Multivariate Analysis?

Univariate analysis studies each biological signal independently. However, the brain is a complex system, so it is often necessary to analyze not only single signals but also their interdependencies.

Multivariate analysis studies how multiple signals interact.

Author note.
A brain region may look unremarkable when studied alone but become important when considered as part of a network. This is why network neuroscience is necessary.

10.2 Network Neuroscience

Network neuroscience aims to map, record, analyze, and model the elements and interactions of neurobiological systems. The objective is to estimate dynamic brain networks at different levels, from molecules and neurons to brain areas and social systems.

10.3 Types of Brain Connectivity

Brain connectivity can be classified into three main types:

1. anatomical connectivity;
2. functional connectivity;
3. effective connectivity.

10.4 Anatomical Connectivity

Anatomical connectivity refers to physical or structural connections linking sets of neurons or neuronal elements.

It is relatively stable at short time scales, such as seconds or minutes, but can change over longer time scales due to plasticity.

It can be measured using:

- invasive tracing studies;
- non-invasive diffusion-weighted imaging techniques.

Anatomical connectivity is the structural basis for functional and effective connectivity, but it cannot fully explain them.

10.5 Functional Connectivity

Functional connectivity describes statistical dependencies between signals. It is model-free and essentially descriptive.

It can be quantified using measures such as:

- correlation;
- coherence;
- transfer entropy.

Author note.
Functional connectivity says that two signals are statistically related. It does not, by itself, tell us whether one signal causes the other.

10.6 Effective Connectivity

Effective connectivity refers to the influence that one neural system exerts over another. It is based on directed or causal influence and usually depends on a model.

It tries to explain observed dependencies in terms of coupling between systems.

Author note.
Functional connectivity is about association. Effective connectivity is about directed influence. This distinction is central in brain network analysis.

10.7 Correlation and Coherence

Correlation measures the linear relationship between two signals in the time domain. Coherence extends this idea to the frequency domain.

Ordinary coherence measures the linear association between two signals at a given frequency.

If $S_{xx}(f)$ and $S_{yy}(f)$ are the power spectral densities of signals x and y, and $S_{xy}(f)$ is their cross-spectral density, ordinary coherence is:

$$C_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)}.$$

10.8 Properties of Ordinary Coherence

Ordinary coherence has the following properties:

- it is a spectral measure, so it depends on frequency;
- it is normalized between 0 and 1;
- $C_{xy}(f_0) = 0$ means independence at frequency f_0;
- $C_{xy}(f_0) = 1$ means maximum linear correlation at frequency f_0.

Author note.
Coherence is useful when we care about rhythms. For example, two EEG channels may be coherent in the alpha band but not in the beta band. This frequency-specific view is lost if we only compute time-domain correlation.

10.9 Limitations of Correlation and Coherence

Correlation and coherence are useful but limited:

- they do not provide directionality;
- they cannot distinguish direct from indirect interactions;
- they may be affected by a common source;
- high correlation does not imply causation.

The common source problem occurs when two measured signals appear correlated because both are driven by a third source.

Author note.
The common source problem is especially important in EEG. Due to volume conduction, the same neural generator can be observed by multiple electrodes, creating apparent connectivity between channels even when there is no true interaction between the underlying brain regions.

11 Brain Networks II: Causality and Directed Connectivity

11.1 Two Definitions of Causality

The slides distinguish two main ideas of causality.

The first is physical influence or control. In this case, changing a cause should change its consequence. This requires controlled interventions and assessment of their effects.

The second is temporal precedence. Causes precede their consequences, so one can test whether past values of one signal improve the prediction of another signal.

Author note.
Physical causality is stronger but harder to test, because it requires intervention. Temporal precedence is easier to test from observational data, but it gives a weaker form of causality.

11.2 Statistical Causality

In the statistical sense, if one can predict a signal better by incorporating past information from another signal than by using only its own past, then the second signal can be considered causal to the first one.

This idea was introduced by Wiener and later formalized by Granger.

11.3 Wiener-Granger Causality

A time series $a[n]$ is said to Granger-cause another time series $b[n]$ if the past of $a[n]$ significantly improves the prediction of $b[n]$ beyond what can be predicted from the past of $b[n]$ alone.

Importantly, Granger causality is directional:

$$a \to b$$

does not necessarily imply:

$$b \to a.$$

11.4 Linear Autoregressive Model

A linear autoregressive model, AR model, predicts the current value of a signal from its past values:

$$x[n] = -\sum_{k=1}^{p} a[k]x[n-k] + e[n],$$

where:

- $x[n]$ is the time series;
- $a[k]$ are autoregressive parameters;

- p is the model order;
- $e[n]$ is the residual or prediction error.

The predicted value is:

$$\hat{x}[n] = -\sum_{k=1}^{p} a[k]x[n-k],$$

and the residual is:

$$e[n] = x[n] - \hat{x}[n].$$

The coefficients are chosen to minimize the squared prediction error.

Author note.
An AR model is basically a memory model: it says that the present value of a signal can be approximated as a weighted combination of its past values.

11.5 Bivariate Autoregressive Model

To test whether x helps predict y, one compares two models.

First, a univariate model predicts $y[n]$ using only the past of y:

$$y[n] = \sum_{k=1}^{p} b_y[k]y[n-k] + e_y[n].$$

Then, a bivariate model predicts $y[n]$ using both the past of y and the past of x:

$$y[n] = \sum_{k=1}^{p} a_{yx}[k]x[n-k] + \sum_{k=1}^{p} b_{yx}[k]y[n-k] + e_{yx}[n].$$

If the second model has a smaller residual variance, then x improves the prediction of y.

11.6 Granger Causality Index

A common Granger causality index from x to y is:

$$GC_{x\to y} = \ln\left(\frac{\mathrm{var}(e_y)}{\mathrm{var}(e_{yx})}\right).$$

If the bivariate model does not improve prediction, then:

$$\mathrm{var}(e_y) \approx \mathrm{var}(e_{yx})$$

and:

$$GC_{x\to y} \approx 0.$$

If the past of x improves the prediction of y, then:

$$\mathrm{var}(e_{yx}) < \mathrm{var}(e_y)$$

and:

$$GC_{x\to y} > 0.$$

Author note.
Granger causality does not mean philosophical causality. It means predictive causality: the past of one signal contains useful information to predict another signal.

11.7 Advantages and Limitations of Granger Causality

Advantages:

- it provides directionality;
- it is based on prediction;
- it can be applied to time series;
- it can be extended to multivariate models.

Limitations:

- it assumes that causes precede effects in time;
- it depends on the quality and stationarity of the data;
- it may be affected by hidden variables;
- pairwise analysis can produce misleading results if other relevant signals are ignored;
- it does not prove physical causation.

11.8 Pairwise and Multivariate Approaches

In the pairwise approach, signals are analyzed two at a time. For example, causality is estimated between x_1 and x_2, then between x_1 and x_3, and so on.

The limitation is that a pair of signals may not contain all relevant information. A third signal may mediate or explain the apparent causal relation.

In the multivariate approach, all signals are modeled together. This can reduce spurious links caused by indirect interactions or hidden common drivers within the measured set.

Author note.
Pairwise Granger causality is easier to compute and interpret, but it can confuse direct and indirect interactions. Multivariate analysis is more realistic for brain networks, but it requires more data and careful model estimation.

11.9 Partial Directed Coherence

Partial Directed Coherence, PDC, is a frequency-domain measure derived from a multivariate autoregressive model. It is used to estimate directed connectivity between signals at different frequencies.

PDC can be used to build directed brain networks where:

- nodes are brain regions or electrodes;
- directed edges represent frequency-specific causal influence;
- edge strength depends on the PDC value.

Author note.
PDC is especially useful when we expect brain interactions to be rhythm-specific. For example, one region may influence another in the alpha band but not in the beta band.

12 Graph Theory and Network Neuroscience

12.1 Graphs and Brain Networks

A graph is a mathematical representation of a network. It consists of:

- vertices, also called nodes;
- edges, representing connections among nodes.

In network neuroscience:

- nodes may represent brain regions or electrodes;
- edges may represent anatomical, functional, or effective connectivity.

Graphs can also be represented using adjacency matrices.

12.2 Graph Properties

Graphs can be classified according to directionality and weight.
According to directionality:

- undirected graphs: edges have no direction;
- directed graphs: edges have a direction.

According to weight:

- binary graphs: edges are either present or absent;
- weighted graphs: edges have numerical values.

12.3 Adjacency Matrix

The adjacency matrix A represents the connections of a graph:

$$A = [a_{ij}].$$

In general:

$$a_{ij} = 0 \quad \text{if there is no link from node } i \text{ to node } j,$$

$$a_{ij} \neq 0 \quad \text{if there is a link from node } i \text{ to node } j.$$

For binary graphs:

$$a_{ij} \in \{0, 1\}.$$

For weighted graphs:

$$a_{ij} = w_{ij}.$$

In undirected graphs, the adjacency matrix is symmetric:

$$a_{ij} = a_{ji}.$$

In directed graphs, the matrix is generally asymmetric.

Author note.
The adjacency matrix is the object from which almost all graph indices are computed. Once you can read the matrix, you can compute degree, density, distance, efficiency, modularity, and other network measures.

12.4 Why Graph Analysis in Neuroscience?

Graph analysis is useful because it provides quantitative indices describing network structure at different scales:

- local scale: properties of single nodes;
- global scale: properties of the whole network;
- mesoscale: communities or modules.

These indices can be used:

- to compare experimental conditions;
- to compare groups of subjects;
- as markers of pathological conditions;
- to support diagnosis, prognosis, or evaluation of treatments;
- as features for classification and decoding.

12.5 Density

The density of a graph measures how many possible connections are actually present.

For a directed graph with N nodes and L directed links:

$$D = \frac{L}{N(N-1)}.$$

For an undirected graph:

$$D = \frac{2L}{N(N-1)}.$$

Density ranges from 0 to 1.

- $D = 0$: no edges;
- $D = 1$: fully connected graph.

12.6 Degree

The degree of a node is the number of its connections.

For an undirected binary graph, the degree of node i is:

$$k_i = \sum_{j=1}^{N} a_{ij}.$$

For a directed graph, one distinguishes:

$$k_i^{in} = \sum_{j=1}^{N} a_{ji},$$

$$k_i^{out} = \sum_{j=1}^{N} a_{ij}.$$

The total degree is:

$$k_i = k_i^{in} + k_i^{out}.$$

Author note.
In brain networks, a node with high degree may be interpreted as a hub, meaning a region that is strongly connected to many other regions.

12.7 Distance

The distance d_{ij} between two nodes is the length of the shortest path connecting node i to node j.

If there is no path from i to j, then:

$$d_{ij} = \infty.$$

Distance is important because it describes how many steps are needed for information to travel from one node to another.

12.8 Global Efficiency

Global efficiency measures how efficiently information is exchanged in the whole network. It is defined as the average reciprocal distance between pairs of nodes.

For a directed graph:

$$E_g = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} \frac{1}{d_{ij}}.$$

For an undirected graph:

$$E_g = \frac{2}{N(N-1)} \sum_{i<j} \frac{1}{d_{ij}}.$$

Global efficiency ranges from 0 to 1:

$$E_g = 1 \quad \text{for a fully connected graph,}$$

$$E_g = 0 \quad \text{for a void graph.}$$

Author note.
Global efficiency is a measure of integration. A network with high global efficiency allows information to travel quickly between distant nodes.

12.9 Local Efficiency

For each node i, one can define a subgraph S_i made of the nodes directly connected to i, excluding i itself. The global efficiency of this subgraph is $E_g(S_i)$.

The average local efficiency is:

$$E_l = \frac{1}{N}\sum_{i=1}^{N} E_g(S_i).$$

Local efficiency measures the tendency of the network to form strongly connected local subgroups. It is also related to robustness: if node i is removed, local efficiency describes how well its neighbors remain connected.

Author note.

Global efficiency is about communication across the whole network. Local efficiency is about how well local neighborhoods remain connected.

12.10 Communities, Divisibility, and Modularity

A community is a subnetwork that shows an organized structure with respect to the entire network. In many applications, it is useful to quantify whether a network can be divided into communities.

Two mesoscale measures are:

- divisibility;
- modularity.

Divisibility measures segregation between communities. Modularity compares the number of intra-community links with the number expected under a random distribution of links.

For undirected graphs, modularity compares:

$$a_{ij} - \frac{g_i g_j}{2L},$$

where:

- a_{ij} indicates the existence or weight of the link between i and j;
- g_i and g_j are the degrees of the two nodes;
- L is the number of links.

If there are more intra-community links than expected by chance:

$$Q > 0.$$

Otherwise:

$$Q \leq 0.$$

For directed graphs, the expected random link depends on in-degree and out-degree:

$$a_{ij} - \frac{g_i^{in} g_j^{out}}{L}.$$

Author note.
Modularity is high when communities are internally dense and externally weakly connected. In brain networks, this is interpreted as network segregation.

12.11 Integration and Segregation

Graph indices can be interpreted in terms of integration and segregation.

A highly integrated network tends to have:

- high global efficiency;
- low local efficiency;
- low divisibility;
- low modularity.

A highly segregated network tends to have:

- low global efficiency;
- high local efficiency;
- high divisibility;
- high modularity.

Author note.
Integration means that distant parts of the network communicate efficiently. Segregation means that specialized communities are strongly internally connected. Healthy brain networks often need a balance between both.

12.12 Reference Networks

Real brain networks can be compared with reference networks.

Main reference networks include:

- regular or lattice networks;
- random networks;
- small-world networks.

A regular network has strong local structure but long paths between distant nodes. It is highly segregated but poorly integrated.

A random network has short paths and high global integration, but poor local structure.

A small-world network combines high local clustering with short global paths. It represents a balance between segregation and integration.

Author note.
The small-world concept is important because brain networks are not expected to be completely regular or completely random. They need specialized local processing, but also efficient global communication.

12.13 Core Exam Takeaways

1. Neuroengineering connects neuroscience, artificial intelligence, robotics, and control engineering.
2. The neuron collects information, integrates it, and generates a binary output.
3. The neuronal membrane is selectively permeable and its electrical behavior depends on ion gradients, channels, and pumps.
4. The Nernst equation gives the equilibrium potential for a single ion family.
5. The resting membrane potential is approximately $-70\,\mathrm{mV}$.
6. Depolarization makes the membrane potential less negative; hyperpolarization makes it more negative.
7. EPSPs depolarize the postsynaptic membrane; IPSPs hyperpolarize it.
8. Temporal and spatial summation integrate multiple synaptic inputs.
9. The action potential is an all-or-none event and represents the binary output of the neuron.
10. Spike timing, not spike shape, carries the relevant information.
11. Continuous conduction is slower; saltatory conduction is faster and depends on myelin and nodes of Ranvier.
12. The cortex is organized in layers, columns, areas, circuits, and networks.
13. Grey matter contains cell bodies and dendrites; white matter contains myelinated axons.
14. The frontal, parietal, occipital, and temporal lobes have different dominant functions.
15. Brodmann areas are based on cytoarchitecture and are associated with function.
16. The visual cortex is retinotopic; the motor and somatosensory cortices are somatotopic.
17. Brain plasticity occurs throughout life and supports learning, memory, recovery, and rehabilitation.
18. The main plasticity mechanisms are synaptic plasticity, axonal sprouting or pruning, axonal regeneration, and unmasking of latent connections.
19. Neural encoding describes how a stimulus is transformed into a neural response.
20. Neural decoding infers the stimulus or intention from the neural response.
21. The rate-coding hypothesis uses firing rate as the main information carrier.
22. A spike train can be represented as a sum of Dirac delta functions.
23. The spike-count firing rate is $r = n/T$.
24. Tuning curves describe the average firing rate as a function of a stimulus feature.
25. A Poisson spike generator models spike counts probabilistically but ignores refractory periods unless constrained.
26. Decoding can be formulated as computing $P(s \mid r)$.

27. Discriminability d' measures how separable two response distributions are.
28. ROC curves describe classifier performance across thresholds.
29. AUC ranges from 0.5, chance level, to 1, perfect classification.
30. Functional connectivity describes statistical dependence; effective connectivity describes directed influence.
31. Coherence measures linear dependence between two signals at a specific frequency.
32. Correlation and coherence do not imply causality and may be affected by common sources.
33. Granger causality is based on improved prediction from past information.
34. Pairwise causality is simple but may be misleading; multivariate causality is more complete.
35. A graph is made of nodes and edges; in neuroscience, nodes may be brain regions or electrodes.
36. Binary and weighted graphs differ in whether edges only indicate presence or also strength.
37. Directed and undirected graphs differ in whether edge direction matters.
38. Density, degree, and distance are basic graph measures.
39. Global efficiency measures network integration.
40. Local efficiency measures local connectedness and robustness.
41. Modularity and divisibility measure community structure and segregation.
42. Small-world networks balance integration and segregation.